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MULTI-SNAPSHOT HNSW INDEX

Abstract

Techniques for generating and leveraging a multi-snapshot HNSW index are provided. In one technique, multiple snapshots of an HNSW index that was generated based on a vector table comprising an initial set of vectors are stored. Each of snapshots has a different build time and is associated with a shared journal that stores a plurality of changes to a plurality of vectors in the vector table. A latest snapshot of the plurality of snapshots is identified. The shared journal is identified. It is determined, based on one or more criteria, whether to generate a new snapshot based on the latest snapshot and the shared journal. In another technique, in response to receiving a vector query that is associated with a timestamp, a particular snapshot is selected from the snapshots based on the timestamp. The vector query is executed against the particular snapshot and the shared journal.

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Background/Summary

BENEFIT CLAIM [0001] This application claims the benefit under 35 U.S.C. § 119(e) of provisional application 63/563,926, filed Mar. 11, 2024, by Lahiri et al., the entire contents of which is hereby incorporated by reference.

RELATED APPLICATIONS

[0002] This application is related to U.S. patent application Ser. No. 18/885,630 filed Sep. 14, 2024; U.S. patent application Ser. No. 18/885,635 filed Sep. 14, 2024; U.S. patent application Ser. No. 18/885,639 filed Sep. 14, 2024; U.S. patent application Ser. No. 18/885,640 filed Sep. 14, 2024; U.S. patent application Ser. No. 18/885,645 filed Sep. 14, 2024 and U.S. patent application Ser. No. 18/828,330 filed Sep. 9, 2024, the contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0003] The present disclosure relates to database management of vectors and, more particularly, to the creation and management of multiple snapshots of an HNSW index.

BACKGROUND

[0004] A vector is a fixed length sequence of numbers, typically floating point numbers, such as [21.4, 45.2, 675.34, 19.4, 83.24], which is a five-dimensional vector. An embedding is a means of representing objects (e.g., text, images, and audio) as points in a continuous vector space where the locations of those points in space are semantically meaningful to one or more machine learning (ML) algorithms. An embedding is often represented as a vector. Generically, a vector embedding represents a point in N-dimensional space. Vector embeddings are intended to capture the important “features” of the data that the vector embeddings represent (or embed). The data a vector embedding represents can be one of many types of data, such as a document, an email, an image, or a video. Examples of features are color, size, category, location, texture, meaning, and concept. Each feature is represented by one or more numbers (dimensions) in the vector embedding.

Hereinafter, a “vector embedding” is referred to as a “vector.”

[0005] Today, vectors are often generated by machine-learned models (e.g., neural networks) and the features they represent are often difficult for humans to understand. One way that vectors are produced by neural networks is by capturing the outputs of the neurons in the penultimate layer, i.e., the neural network's outputs just before the final processing layer.

Distance Between Vectors

[0006] An important attribute of vectors is that the distance between two vectors is a good proxy for the similarity of the objects represented by the vectors. Two vectors that represent similar data should be a short distance from each other in vector space. The opposite is also true: dissimilar data are represented by vectors that are far apart from each other in the vector space. For example, the distance between a vector for the word “cat” and a vector for the word “dog” should be less than the distance the vector for the word “cat” and a vector for the word “plant.”

[0007] The distance between two vectors is often calculated by summing the squares of the difference between the numbers in each position of the vectors:

[00001](Vector1[1] - Vector2[1])² + (Vector1[2] - Vector2[2])² + . . . Math.

[0008] The property that vector distance represents object similarity is what allows similar data to be found using a vector database. For example, when a vector representing a picture of a dog is searched for in a vector database, the nearest vectors will be those representing other dogs, not vectors representing plants.

Vector Processing Workloads

[0009] Vector processing workloads (not to be confused with SIMD vector processing) have been used in Natural Language Processing (NLP), image recognition, recommendations, etc. Vector processing workloads have two sub-categories that require separate optimization strategies: indexing and searching. Regarding indexing, vector embeddings (or simply vectors) are indexed using approximate indexing techniques. Unlike B-tree indexes, a vector index returns many matching values ranked by similarity. Index creation and rebuild tend to be CPU intensive and are optimized for throughput.

[0010] Regarding searching, the stored vectors are searched using a class of algorithms known as “Similarity Search” or “Approximate Nearest Neighbor (ANN)” to find the closest vectors to a query vector. Search is designed to minimize CPU usage in order to minimize response time.

Vector Processing Patterns

[0011] A vector similarity search is like interactive online transaction processing (OLTP) in that end-users submit vector queries and expect an instant reply. Vector similarity search requires millisecond response time to finding vectors that are close (represent similar data) even when the database in which the vectors are stored holds billions of vectors. An example query is “find products that are similar to this picture” [reference to a digital image].” Another example query is “find corporate documents that conceptually match this natural language prompt: [NL prompt].”

[0012] Providing fast response times requires using specialized vector indexes and fast algorithms for computing distances between vectors. In some use cases, there is a need to combine vector similarity search with relational data. For example, a query may ask for data about houses that match a natural language prompt, are valued at over \$1 M, are in zip code 94070, and whose owner recently declared bankruptcy. Also, there may be a need to be able to insert new vectors into a database, delete vectors from the database, and index the vectors in real time.

Vector Databases

[0013] Early vector workloads often used flat files or object stores to store vectors. An application would read the vectors out of their backend repositories into memory and perform vector processing using third-party libraries, such as FAISS. Generative artificial intelligence (AI) has greatly increased the volume and processing needs for vectors. Generative AI requires support for much higher volume ingest and faster filtering and retrieval. A database with vector capabilities and built-in indexing is important for these applications.

[0014] The approaches described in this section are approaches that could be pursued, but not necessarily approaches that have been previously conceived or pursued. Therefore, unless otherwise indicated, it should not be assumed that any of the approaches described in this section qualify as prior art merely by virtue of their inclusion in this section.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] In the drawings:

[0016] FIG. 1 is a block diagram that depicts an example vector database management system (VDBMS), in an embodiment;

[0017] FIG. 2 depicts an example vector object, in an embodiment;

[0018] FIG. 3 is an example process for retrieving a version of a vector from a vector object, in an embodiment;

[0019] FIG. **4** is a flow diagram that depicts an example process for storing multiple versions of a vector into a single vector object, in an embodiment;

[0020] FIG. **5** is a diagram that depicts an example row of a table with a vector column that contains references to versions of a vector, in an embodiment;

[0021] FIG. **6** is a diagram that depicts an example set of five clusters that is generated based on a clustering algorithm, in an embodiment;

[0022] FIG. **7** is a diagram that depicts a subset of clusters that are selected based on distance calculations between query vector and centroids in clusters;

[0023] FIG. **8** is a diagram that depicts an example logical representation of an HNSW index **800** that comprises four layers, in an embodiment;

[0024] FIG. **9** is a block diagram that depicts an example storage architecture for storing an HNSW graph, in an embodiment;

[0025] FIG. **10** is a block diagram that depicts an example organization of an example HNSW index, in an embodiment;

[0026] FIG. **11** is a block diagram that depicts an updated version of HNSW after multiple inserts have been made into that index, in an embodiment;

[0027] FIG. **12** is a flow diagram that depicts an example process for recording changes to vectors and leveraging those changes during processing of a vector query, in an embodiment;

[0028] FIG. **13** is a flow diagram that depicts an example process for processing an in-transaction read associated with a vector query, in an embodiment;

[0029] FIG. **14** is a flow diagram that depicts an example process for creating and maintaining multiple snapshots of an HNSW index, in an embodiment;

[0030] FIG. **15** is a flow diagram that depicts an example process for creating a new snapshot of an HNSW index, in an embodiment.

[0031] FIG. **16** is a block diagram that illustrates a computer system upon which an embodiment of the invention may be implemented;

[0032] FIG. **17** is a block diagram of a basic software system that may be employed for controlling the operation of the computer system.

DETAILED DESCRIPTION

[0033] In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be apparent, however, that the present invention may be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring the present invention.

General Overview

[0034] A system and method for creating and maintaining multiple snapshots of an HNSW index. In one technique, multiple snapshots of an HNSW index are stored. A vector query that is associated with a timestamp is received. Based on the timestamp, the most recent snapshot, of the multiple snapshots, that has an earlier build time than the timestamp of the vector query is selected. A shared journal (which may be implemented as a table) that is associated with the HNSW index is identified. The shared journal tracks changes after the initial creation of the HNSW index. The vector query is executed against both the selected snapshot and the identified shared journal. Results from both executions are combined to generate a final result of the vector query.

Vector Database System Overview

[0035] FIG. **1** is a block diagram that depicts an example vector database management system (VDBMS) **100**, in an embodiment. VDBMS **100** comprises a vector database server **110** and a vector database **120**. Vector database server **110** is communicatively coupled to vector database **120**. VDBMS **100** may be deployed in a network of an enterprise or may be deployed in a cloud environment and, therefore, may be accessible to an enterprise over one or more computer networks (e.g., the Internet). VDBMS **100** may be provisioned for an enterprise by a cloud

management team of a cloud provider as needed on an enterprise-by-enterprise basis.

[0036] Vector database server **110** comprises one or more computing machines, each executing one or more compute instances that receive and process data requests, including data retrieval requests (e.g., queries) and data modification requests (i.e., for vector data modifications), such as inserting vectors, deleting vectors, and updating vectors. A computing instance translates a data request into a storage layer request that the computing instance transmits to vector database **120**. A computing machine that hosts at least one compute instance includes (1) one or more processors, (2) volatile memory for storing data requests (and their respective contents) and vector data that is retrieved from vector database **120**, and (3) optionally, non-volatile memory.

[0037] Vector database **120** may comprise multiple storage devices, each storing vector data and, optionally, one or more non-vector data. For example, vector database **120** stores a table that includes a column for storing vectors and one or more column for storing user data, such as a column for storing a user identifier, a column for storing a user profile, a column for storing user search history, a column for storing user access history, a column for storing user-generated content, etc. In this example, each row in the table corresponds to a user, such as a customer, a subscriber to a service, etc.

[0038] Vector database **120** may also store one or more indexes that index content in vector database **120**, such as content stored in one or more base tables. Some of the indexed content may be vector-related data (e.g., actual vector embeddings and metadata thereof) and some of the indexed content may be non-vector-related data, such as content in columns that do not store vectors. Thus, at least one index that vector database **120** may store is a vector index, described in more detail herein.

Native Support for Vector Embeddings

[0039] In an embodiment, VDBMS **100** provides native support for a new vector embedding (“VECTOR”) datatype. The native support may include operators and indexes that are associated with that datatype. Examples of other datatypes that may be natively supported by VDBMS **100** include INT (integer), FLOAT (floating point), DATE, and STRING.

[0040] A VECTOR column may be defined with two values: a number of elements/dimensions and a dimension format (or “element type”). A generic example of VECTOR datatype syntax is the following:

TABLE-US-00001 VECTOR(<Num Elements>, <Element Type>). The following is an example is using this VECTOR datatype syntax to create a new table for storing vectors: create table vector_tab (id NUMBER, attributes JSON, data VECTOR(768, ‘FLOAT32’)) where “id”, “attributes”, and “data” are names of columns of the table named “vector_tab.” The first column in vector_tab is a NUMBER datatype, the second column in vector_tab is a JSON datatype, and the third column in vector_tab is a VECTOR datatype.

[0041] Different types of functions may be supported as part of this new VECTOR datatype. Example functions include distance functions, aggregate functions, and single vector functions. An example distance function is “VECTOR_DISTANCE(vector1, vector2, <optional distance metric>).” This distance function computes a distance between vector1 and vector2 and is the most common operation forming the basis of similarity search. Distance metrics may be in Euclidean (which may be the default metric), cosine distance (1–cosine similarity), dot distance (negative of dot product), Manhattan, Hamming, etc. This distance function may return different types of values depending on the storage representation. For example, this distance function returns (a) a binary float value if the storage representation is four bytes or less or (b) a binary double value otherwise.

[0042] An example of an aggregate function is “VECTOR_AVG(VECTOR)” where VECTOR refers to the column and, thus, this function takes a set of vectors as input. This function computes the average vector across a set of vectors and returns a vector. This function is useful for Word2Vec use-cases (e.g., sentiment analysis across tweets) where every word has a vector and a sentence's vector is computed as the average vector of all words in the sentence or all keywords in the

sentence.

[0043] Regarding single vector functions, two example functions are as follows: (1)

“VECTOR_COUNT_DIMENSIONS(VECTOR),” which counts the number of dimensions in an input vector and returns a number; and (2) “VECTOR_NORM(VECTOR),” which computes the Euclidean norm/length of an input vector and returns a value, such as a BINARY DOUBLE.

[0044] The following are examples of queries (in structure query language (SQL) format) using the vector distance function:

TABLE-US-00002 select id from vector_tab order by VECTOR_DISTANCE(data, :query) asc
fetch first 5 rows only; This query compares the query vector (“:query”) with every vector in
the table “tab” (where tab has a column named “data” that stores vectors). While a result of this
query is 100% accurate, it is relatively slow. select id from tab t where t.attributes.year.number()
< 2000 order by VECTOR_DISTANCE(data, :query) asc fetch first 5 rows only; This query
results in obtaining the top five photos that are similar to a query photo (“:query”) that were taken
before the year 2000. with query as (select id, data from vector_tab where id = :id)
select t.id from vector_tab t, query q where t.id != q.id order by VECTOR_DISTANCE(t.data,
q.data) asc fetch first 5 rows only; This query results in obtaining the top five nearest neighbors to a
specific vector in a data set (i.e., vector_tab). select id from vector_tab where
VECTOR_DISTANCE(data, :query, ‘MANHATTAN’) < 5; This query results in obtaining all
neighbors that are within a threshold distance from the query vector (“:query”), where the vector
distance function specifies a specific distance metric (i.e., Manhattan), thus, overriding the default
distance metric.

Vector Storage

[0045] Because vectors that are produced by a ML model are of fixed length, an optimal type is used for underlying storage. For example, for vectors that are less than 8K elements in length, RAW may be used, which should handle most use cases. For larger vectors, binary large object (BLOB) may be used. BLOB should not be used for small vectors due to fixed overhead of LOBs. However, whether BLOB or RAW is used has no effect on the user interface.

[0046] In an embodiment, additional summary data may be kept within a vector in order to accelerate operations. For example, the squared norm of a vector (Sum(v.sup.2)), which is required for distance calculations, may be stored in a header of the vector. A vector may require additional metadata, such as vector version number, whether the vector stores IEEE floats or binary floats, etc.

Vector Datatype: Flexible Type Specification

[0047] In an embodiment, the VECTOR datatype is specified with a flexible dimension count and/or a flexible dimension format. Supported dimension formats may include INT8 (1-byte integer), BINARY, FLOAT16, FLOAT32, FLOAT64, and BFLOAT16. An example of a support VECTOR datatype specification is the following: [0048] VECTOR or VECTOR(*, *)

[0049] In the above example, both the dimension count and the dimension format are flexible. This allows the greatest flexibility. In this way, if another vector with a different dimension count or different dimension format is generated, that vector may be stored with other vectors of different counts and/or dimensions, without having to change any schema or applications that target vectors that are defined accordingly. [0050] VECTOR(<dimension count>) or VECTOR(<dimension count>, *)

[0051] In the above example, the dimension count is fixed, but the dimension format is flexible.

[0052] VECTOR(*, <dimension format>)

[0053] In the above example, the dimension count is flexible (and could theoretically be any value), but the dimension format is fixed. [0054] VECTOR(<dimension count>, <dimension format>)

[0055] In the above example, both the dimension count and the dimension format are fixed. A specific example of using the VECTOR data type when creating a table is the following: [0056] create table vectab (id number, c VECTOR(1024, FLOAT32))

where c is the name of a column of datatype VECTOR.

[0057] An advantage of a flexible specification is that API calls are easier, since only the VECTOR name needs to be passed without having to specify number of dimensions and/or dimension format.

[0058] Another advantage of a flexible specification is that it allows a user (e.g., a database administrator) to evolve the contents of a VECTOR column over time easily. There are a wide range of embedding models with different dimension counts and dimension formats that can be chosen to vectorize user data. For example, Open AI Text-Ada-002 produces vectors of 1536 dimensions of FLOAT32, Cohere Embed-English-v3.0 produces vectors of 1024 dimensions of FLOAT32, and Alibaba's gte-small-ct2-int8 produces vectors of 384 dimensions of INT8. A user may desire to try out vectors from various models and judge the quality of semantic search results before finalizing a model. Having a flexible specification allows a user to keep the schema consistent while changing the content stored in the vector column.

[0059] Often the user may choose to partition the data in a table by some relational attributes and each partition can contain vectors of different dimension counts or formats. For example, a user may choose to partition a BOOKS table by the GENRE column. Certain genres like Fiction or Economics might be more popular than genres like Biography. Books of the more popular genres can be vectorized using higher dimension vectors while less popular genres can be vectorized with lower dimension vectors. Using higher dimension vectors to improve searches implicitly assumes that higher dimensional vectors capture more “semantic information.” Thus, higher dimension vectors may be used to find matches to a wider array of user searches.

[0060] A disadvantage of having flexible dimension counts is that vector distance computations cannot be blindly performed on a column containing vectors of different dimension counts. For example, in the book genre example above, a user must add a predicate on the GENRE column to ensure that the search vector is being compared with vectors of the same dimension count.

[0061] However, vector distance operations may be executed on two vectors of different dimension formats. For example, using a new SQL function VECTOR_DISTANCE(), a distance computation may be performed between a three-dimensional vector of FLOAT32 and a three-dimensional vector of FLOAT64. The following is an example of a data definition language (DDL) statement, a data manipulation language (DML) statement, and a structure query language (SQL) statement, respectively:

```
TABLE-US-00003 create table vectab (c1 vector(3, FLOAT32), c2 vector(3, FLOAT64)); insert into vectab values (TO_VECTOR('[1.15, 2.27, 3.34]', 3, FLOAT32), TO_VECTOR('[1.234, 2.234, 3.334]', 3, FLOAT64)); select vector_distance(c1, c2) from vectab;
```

[0062] Internally, vectors with the lower precision dimension format are upconverted to the higher precision format and then the distance computation is performed. This ability adds to the advantages of flexibility described earlier.

Vector Datatype: Flexible Storage Format

[0063] In an embodiment, vectors are stored in objects, such as large objects (LOBs), an example of which is a binary LOB (BLOB). Storing vectors in LOBs allows for storing large vectors, such as vectors with dimensions up to 65,534 dimensions.

[0064] In an embodiment, an object that stores a vector is designed to be self-contained, meaning each vector object contains information about the dimension count and/or dimension format of the corresponding vector. This allows any module or application to examine a vector object and precisely interpret the vector without relying on a separate dictionary/catalog table that describes the datatype.

[0065] In a related embodiment, the format of a vector is designed to cache additional metadata that can be used to accelerate distance computations during run-time. One such metadata is the Squared L2-Norm (Euclidean Norm) of a vector, which norm can be used to speed up Euclidean distance calculations. Given two vectors v1: (x1, y1) and v2: (x2, y2), the Euclidean distance between the two vectors is $\sqrt{(x1-x2)^2+(y1-y2)^2}$.

The portion inside the $\text{sqrt}()$ can be expanded as: $x_1^2 + x_2^2 - 2x_1x_2 + y_1^2 + y_2^2 - 2y_1y_2 = (x_1^2 + y_1^2) + (x_2^2 + y_2^2) - 2(x_1x_2 + y_1y_2) = \text{Squared_Norm}(v_1) + \text{Squared_Norm}(v_2) - 2(x_1x_2 + y_1y_2)$. Thus, for a given query vector, a one-time computation of the Squared Norm may be computed and, if each vector in a table already has the Squared Norm cached, then the distance computation cost is reduced by approximately 10%.

[0066] The format of a vector is designed to store the vector's data, including floating point dimension formats, in either IEEE754 format or a proprietary canonical binary float/double format that allows for floating point numbers to be byte-comparable. An example vector format is as follows:

TABLE-US-00004 [Version # (1B)][Flag (2B)][Num_Dims (1B/2B/4B)][Storage Type (1B)][Squared L2-Norm (1B/2B/4B/8B)][Vector Data (1B/2B/4B/8B)*num_dims]
 where each '[]' corresponds to a field in a vector, 'B' refers byte, '1B/2B' means that the corresponding field in a vector may be one byte or two bytes in length, 'Num_Dims' refers to number of dimensions, example storage types include FLOAT32 and INT8, and L2-Norm is the Euclidean distance between the vector and the "zero" vector (or origin). Calculating and storing the L2-Norm value within a vector object reduces the time to compute a distance between the vector and another vector.

Versioning of Vectors

[0067] An important piece of an AI Vector Search eco-system is the ability to update the embedding model that is used to vectorize (e.g., unstructured) data. As this space is rapidly evolving, it is possible that embedding models of the future produce vectors of different dimension counts and/or different dimension formats. Having a flexible VECTOR column type allows users to update the vector column by replacing vectors from an old model with a new model. While functionally valid, this approach may prove to be expensive, especially for large datasets with hundreds of millions of vectors. In particular, all vectors in the table for a specific column must be updated with vectors from the new model in a single transaction before new searches can leverage them. Such an update could take hours.

[0068] Also, users may want to experiment with different embedding models to decide which model's vectors provide the best semantic search quality. One idea is to create multiple columns, one for each embedding model. However, this requires creating multiple versions of the application that references different vector columns in Top K queries.

[0069] In an embodiment, multiple versions of a vector are stored in the same object. Storing multiple versions of a vector in the same object addresses problems of both approaches (of (i) replacing old vectors with new vectors and (ii) creating a column for each embedding model). An object may be a LOB (or large object), an example of which is a binary LOB (BLOB). Relatively small vectors may also be stored in RAW columns.

[0070] From a storage perspective, there are two main options to store multiple versions of a vector in a single object (e.g., a BLOB): (1) storing the different versions in a linked-list style format within the BLOB format and (2) leveraging vector-only extents. Regarding the first option, FIG. 2 depicts an example vector object **200**, in an embodiment. Vector object **200** comprises two versions (**210** and **220**) of a vector. Data for each version comprises three fields: a model version number, a next version reference that points to the next version in the same object, and the vector data itself of that version. Therefore, data for the first version **210** comprises model version number **212**, next version reference **214** (which references version **220**), and vector data **216**. Similarly, data for the second version **220** comprises model version number **222**, next version reference **224**, and vector data **226**. Vector data **216** stores the embedding data for version **210** while vector data **226** stores the embedding data for version **220**.

[0071] Thus, a next version reference field stores a value that indicates a location where data about

another version is stored. The value may be a byte offset into the version object. The last (in order) version in a vector object may have a value of zero or null in its next version reference field, which value indicates that there are no more versions of the vector that follow that last version.

[0072] The model version number of a version may be a number that is automatically set by the process (e.g., a version adding component) that inserts the version into a vector object. The model version number may be a monotonically increasing value. For example, the first version of a vector is assigned model version '0', the second version of the vector is assigned model version '1' and so forth. Alternatively, the model version number may be (a) a value that corresponds to the embedding model that generated the version or (b) a name of that embedding model. In either scenario, vector database **120** stores a mapping (which may be read into memory of vector database server **110**) that maps embedding model names/identifiers to their respective model version numbers that are stored in vector objects in vector database **120**.

Versioning of Vectors: Retrieving a Version

[0073] A version retrieval component retrieves one or more versions of a vector from a vector object. The version retrieval component may be implemented in software, hardware, or any combination of software and hardware. The version retrieval component may be part of a vector database server **110** and may be called by a vector search application. Alternatively, the version retrieval component may be part of a storage sub-layer that is distinct from a database server layer that receives vector search queries. For example, the version retrieval component may be part of vector database **120**. The more processing that is pushed to vector database **120**, the less data that needs to be transferred to vector database server **110**.

[0074] The version retrieval component may determine which version(s) to retrieve based on one or more inputs (e.g., from a user or a vector search application). For example, a user specifies which version of a vector is desired, such as "version0," "version1," "3," etc. A version specification may be passed as input to an application that processes versioned vectors.

[0075] Additionally or alternatively, the version retrieval component retrieves the most recent version of a vector by default. Thus, no input specifying which version(s) to retrieve may be necessary. In this way, if a user/application does not specify a version number, then it is presumed that the user/application desires the most recent version.

[0076] In a related embodiment, new SQL syntax is provided to allow users to specify which version of a vector is desired. For example, the vector distance function may be augmented to allow for flexible version specification, such as the following:

TABLE-US-00005 VECTOR_DISTANCE(<vec_col1>, <vec_col2>, <distance metric>, <version number of vec_col1>, <version number of vec_col2>)

The version numbers may be bind values that an application can change.

[0077] FIG. **3** is an example process **300** for retrieving a version of a vector from a vector object, in an embodiment. Process **300** may be performed by a version retrieval component.

[0078] At block **310**, a particular version of a vector is identified. The particular version may be specified by a user. Alternatively, the particular version may be a default version, such as the oldest version or the newest (or most recent) version. Different API calls to the version retrieval component may indicate which version. For example, one API call may be associated with a request to retrieve the oldest version of a vector while another API call may be associated with a request to retrieve the newest version of a vector. Alternatively, only a single API call is used to initiate the version retrieval component and one or more values that are passed as part of the API call indicate which version(s) to retrieve.

[0079] At block **320**, a vector object is selected. Block **320** may involve selecting multiple vector objects. A vector object may be selected based on applying one or more search criteria to one or more columns of a table that stores vectors. Block **320** may be performed before or after block **310**.

[0080] At block **330**, a version in the vector object is identified. The first iteration of block **330** may involve identifying the first version (sequentially speaking) in the vector object. The second

iteration of block **330** may involve identifying the next version in the vector object, which is after the first version, using the next version reference field value identified in block **360**.

[0081] At block **340**, it is determined whether the identified version in the vector object corresponds to the particular version that was identified in block **310**. Block **340** may involve comparing the particular version (identified in block **310**) with the value in the model version field of the identified version. If the determination in block **340** is in the affirmative, then process **300** proceeds to block **350**. Otherwise, process **300** proceeds to block **360**.

[0082] At block **350**, the version data of the identified version is retrieved from the version object. The version data may be identified based on (1) first data that indicates an offset into the vector object where the version data begins and (2) second data that indicates a length (e.g., in bytes) of the version data. Thus, the vector data between (a) a first location indicated by the first data and (b) a second location indicated by a combination of the first data and the second data (e.g., first data+second data) is retrieved.

[0083] After block **350**, if multiple versions of the vector are requested, then process **300** may return to block **310** where another version is identified. For example, a request to the version retrieval component may specify versions 4 and 5 or the second version and the most recent version. Therefore, process **300** may be performed once for each requested version.

[0084] At block **360**, a location of the next vector data within the vector object is identified. This location may be identified using the value in the next version reference field of the identified version (identified in block **330**). Process **300** then returns to block **330**. However, if the value in the next version reference field indicates that there are no versions, then process **300** may return an error or return a value indicating that the particular version is not available.

[0085] In a scenario where the version retrieval component receives a request to identify and return the most recent version of a vector from a vector object and the most recent version is stored at the beginning of the vector object at a position that is known without having to scan the vector object (e.g., because the first version in a vector object is always stored at offset six bytes from the beginning of the vector object), then identifying the most recent version involves identifying that byte offset into the vector object and returning the bytes between (1) the byte offset and (2) a location identified by the sum of (i) the byte offset and (ii) the length of the vector data of the most recent version.

Versioning of Vectors: Deleting Particular Versions

[0086] In an embodiment, a request or instruction is received to delete a particular version of multiple vectors. For example, a software engineer may decide that the embedding model that generated the most recent versions of a set of vectors performed poorly in one or more tests. In order to free up space in non-volatile, or persistent, storage, the most recent versions of the set of vectors are deleted.

[0087] Deleting a particular version of a multiple vectors may involve receiving an instruction that indicates the particular version (e.g., the first version or the most recent version or a value that indicates a particular number). The instruction may also specify a table or a column within the table that stores the vectors. In this way, the set of vectors involved may be inferred. For each vector object in the table or column, a version deletion component (e.g., of vector database server **110** or of vector database **120**) determines the version in the vector object (logical or physical, which is described in more detail herein) that matches the particular version and either (i) deletes the embedding data of that version or (ii) sets a flag that indicates that the space occupied by the embedding data is reusable. If future versions are expected, then approach (ii) may be preferred since the space has already been allocated.

[0088] In a related embodiment, an instruction or request to delete may specify or otherwise indicate multiple versions. For example, an instruction may be to delete versions two and four, or to delete the last two versions.

Versioning of Vectors: Adding New Versions

[0089] In an embodiment, adding a new version to a vector object that comprises one or more versions involves appending the new version to the one or more versions. Such adding is efficient with relatively low overhead. Adding a new version may involve traversing one or more next version reference fields in a vector object. For example, once a version object is identified, a version adding component identifies a next version reference field in the data for the first version (sequentially) in the version object. The version adding component uses the value in the next version reference field to identify the second version (sequentially) in the vector object, if the second version exists. This process continues until the version adding component identifies, within the vector object, a next version reference field that contains a value that indicates that there are no more versions in the vector object. The version adding component identifies a position within the vector object to which the new version may be added. That position is the byte that follows the last byte of vector data in the vector object.

[0090] In adding a new version of a vector to a vector object, the version adding component also adds a value to a model version number field for the new version, the value indicating the version of a model that generated the new version. The version adding component may also add a value for the next version reference field. When appending a new version to a version object, this value may be zero or null, indicating that there are no more versions after this new version.

[0091] However, appending versions to one or more versions in a vector object may cause (due to data block size limitations) the most recent version to be stored in a different data block than the data block that stores the one or more versions. Therefore, when retrieving the most recent version, a version retrieval component must follow one or more references to arrive at the different data block to retrieve that version. Accessing two or more data blocks to retrieve the proper version of a vector may increase latency substantially. Fitting as many versions of a vector into a single data block is preferable for use cases where there is significant traversing of versions involved.

[0092] In a related embodiment, a vector object includes a most recent reference field that includes a reference or pointer to the most recent version of the vector represented by the vector object. This most recent reference field may be the first field in the vector object or one of the first few fields in the vector object, which field may be easily and quickly identifiable, such as N bytes from the beginning of the vector object. In this way, retrieving the most recent version of a vector may only require following at most one reference, even though there may be many versions of the vector that are stored in the vector object.

[0093] In another embodiment, a new version is prepended to a vector object that comprises one or more versions. Such prepending may require shifting existing contents of the vector object to later offsets or positions within the vector object. Shifting may comprise copying existing contents (e.g., vector data of multiple versions of a vector, model version numbers, and next version reference values) of the vector object, determining a byte offset in the vector object, and storing the copied contents beginning at the byte offset, whether in the same data block or a new data block.

Determining the byte offset may involve determining the size (e.g., in bytes) of the new version, determining the size (e.g., in bytes) of any required fields that are to accompany the new version (such as a model version number field and a next version reference field) and totaling/summing those two sizes to compute the byte offset.

[0094] Prepending a new version to a vector object also involves generating a value for a next version reference field of the new version and storing that value in that next version reference field. The value in this next version reference field points to the most recent version (before the new version) that was added to the vector object. The first time a version of a vector is stored in the vector object, the value of the next version reference field may be zero or null, indicating that there are no more versions sequentially after the first version is added to the vector object. Thereafter, the value of the next version reference field for the new version may be the size of the new version plus zero or more pre-defined offsets.

[0095] Similarly, for inserting vectors, new DDL may be used to specify the version into which to

insert the vectors. There are at least two techniques to specify which version is to be updated or retrieved. A first technique is to use a SQL construct to specify which version is of interest. For example, in order to update a value of the fourth version of a vector using JSON-like interpretation, the following statement may be used: [0096] update mytab set
 veccol=VECTOR_TRANSFORM(veccol, SET ‘\$[4]’=[1.1, 2.2, . . .]’)

As another example, in order to retrieve the third version using JSON-like interpretation, the following statement may be used: [0097] select FROM_VECTOR(VECTOR_VERSION(veccol, ‘\$[3]’)) from mytab;

[0098] A second technique is to create a DDL that defines the currently accepted version across sessions. For example, in order to make the third version the default version for any application that accesses the corresponding column, the following statement may be used: [0099] alter table mytab modify column (veccol current_version 3);

With this added metadata associated with the vector column, when inserting vectors into that column, the following statement may be used to automatically update vector column payloads to add the third version: [0100] insert into mytab values (‘[3.1, 3.2, 1.0]’);

[0101] The alter table statement above works with reads and writes. Thus, queries on this column will read the third version. The following alter table statements are specifically only for reads and writes, respectively: [0102] alter table mytab modify column (veccol current_read_version 3); [0103] alter table mytab modify column (veccol current_write_version 3);

[0104] If an alter table statement sets the version to 3 and there are no versions 1 and 2, then a subsequent insert may simply insert the new payload into version 3 and have implicit NULLs for versions 1 and 2. Alternatively, an error may be thrown in case a vector has no explicit values for versions 1 and 2.

[0105] An advantage with this embodiment is that the new vectors can be added in a rolling/online fashion where the application can continue to use old vectors while the new vectors are added over time. For example, a vector column has version 1 for all rows. An application is running searches against version 1. Then, a user (e.g., database administrator (DBA)) decides to try out a new embedding model and creates version 2 for all vectors. A parallel session can now generate vectors that get added as version 2 to every row. This can be done in batches of, for example, 100 rows, such that it is not an all-or-nothing decision. The entire table can get version 2 vectors across many transactions. The online nature is highlighted by the fact that the existing application can continue reading version 1 for its searches, even though some (or all rows) have version 2 added. Further, other non-vector transactions can happen concurrently since version 2 can be generated in batches.

[0106] Contrast this with a solution where the VECTOR column is not versioned and the DBA must update a non-versioned vector, i.e., all version is must be replaced with version 2s. This can be done with an update statement. If the update statement updates all rows in the dataset in a single transaction, then a mechanism of snapshot isolation provides a similar online behavior. All new versions will be part of a single active transaction. Queries will roll it back as part of Consistent Read image construction (because the transaction is active) and can read version 1. (This may be a standard mechanism for allowing reads while DMLs are in progress). Note that this only works if all rows are updated in the same transaction. If this is done in batches, then a situation may arise where rows 1 to 1 M (one million) have version 2 vectors and rows 1 M to 1B (one billion) have version 1 vectors, resulting in queries having wrong results. However, updating a 1B row table in a single transaction is not conducive for many reasons. The instance could crash before all 1B rows can be updated leading to wasted work. Embedding generation is expensive, both from a time and money perspective. Thus, re-generating the embeddings is not efficient. The transaction needs to keep row locks on modified rows which means no other transactions can modify the same rows (for example, to change a different relational column). Since the entire transaction could be very long, other writes may be stalled in the system, even though reads are fine.

[0107] FIG. 4 is a flow diagram that depicts an example process **400** for storing multiple versions of a vector into a single vector object, in an embodiment. Process **400** may be performed, at least in part, by the version adding component, which may be part of vector database server **110** or vector database **120**.

[0108] At block **410**, a first version of a vector is stored in a vector object. The first version may have been generated by a first embedding model (e.g., a neural network) and stored in a row of a table with a column for storing objects of the VECTOR datatype. The vector object may be a BLOB object.

[0109] At block **420**, a second version, of the vector, is identified. The second version is different than the first version and is not yet stored in the vector object. The second version may have been generated by a second embedding model after the vector object was created and after the first version of the vector was stored in the vector object. Block **420** may involve identifying the second version immediately after the second version is generated.

[0110] At block **430**, an instruction to store the second version in the vector object is received. The instruction may have originated from a storage application that transmitted the instruction to a vector database server, such as a vector database server **110**. Alternatively, the instruction may originate from the vector database server and be received at a storage layer of vector database **120**.

[0111] At block **440**, in response to receiving the instruction, the vector object is identified. The instruction may include a row identifier that uniquely identifies a row in which the vector object is stored. Alternatively, the instruction may include other data (such as a combination of data values) that is used to uniquely identify a row.

[0112] At block **450**, the vector object is updated to include the second version in addition to the first version. Block **450** may involve appending, within the vector object, the second version to the first version. Alternatively, block **450** may involve prepending the second version to the first version.

[0113] At block **460**, a value that indicates a location, within the vector object, of the first version or of the second version is inserted into a next version reference field of the vector object. Block **460** may be part of block **450** in that, during the update, other data may be inserted into the vector object. Other data may include this value for a next version reference field, as well as a model version number (or identifier) that identifies (and/or is mapped to) an embedding model that generated the second version.

Vector-Only Extents: Logical Vector Objects

[0114] As noted above, there are two main options to store multiple versions of a vector. FIG. 2 depicts a first main option (i.e., a linked-list style format) while vector-only extents are a second main option. An extent is a logical unit of database storage space allocation made up of a number of contiguous data blocks.

[0115] In an embodiment, the versions of a vector object may be physically stored in one or more vector extents, where a vector-only extent only contains vector data, including a vector embedding of a version of a vector. Thus, each vector version is stored in blocks allocated for vector-only extents. A vector-only extent might store vector versions from different vector objects. For example, vector-only extent E1 stores {Vector Object #1, Version #1}, {Vector Object #1, Version 2}, {Vector Object #2, Version 2}, and another extent E2 stores {Vector Object #1, Version 3}, {Vector Object #2, Version 1}, {Vector Object #2, Version 3}.

[0116] FIG. 5 is a diagram that depicts an example row **500** of a table with a vector column **520** that contains references **522-526** to versions of a vector, in an embodiment. Row **500** includes a first column **510** (e.g., a name), vector column **520** (which is the second column in the table), and a third column **530** (e.g., an employment start date). The contents of first column **510** are stored in first column **510**, the contents of third column **530** are stored in third column **530**, but the vector embeddings associated with vector column **520** are stored in one or more vector objects (or “vector-only extents”) that are stored separate from vector column **520**. Instead, vector column **520**

stores version references **522-526**, each referencing a separate version of a vector. Each separate version of the vector is stored in a different (physical) vector-only extent, i.e., vector-only extents **542-546**. Thus, a single row in the table comprises contents from first column **510**, a set of versions references, and contents from third column **530**.

[0117] A single vector-only extent may store multiple vector embeddings of one or more vectors. For example, the first two versions of a vector are stored in one vector-only extent, while a third version of the vector is stored in a different vector-only extent.

[0118] In this embodiment, vector column **520** does not physically contain vector embedding data (which is stored in vector-only extents **542-546**), only non-vector embedding data, such as version references **522-526**. Each version reference may be the same fixed size.

[0119] In a related embodiment, vector column **520** also stores a model version number for each version reference. These model version numbers may be used by a version retrieval component to identify the requested version of a vector. Similar to the process above for adding new versions of a vector to a vector object, when new versions are added to a logical vector object, version references associated with the versions may be appended to one or more existing version references in the logical vector object, prepended to the one or more existing version references, or added using a different technique.

[0120] In this embodiment where a version column contains version references instead of the actual vector embeddings, because many version references may fit into a single column, a version retrieval component must follow at most a single version reference to retrieve a vector-only extent that contains the vector embedding data for a requested version of a vector. When identifying a particular version of a vector, the process that is attempting to read the particular version may scan each version reference sequentially in order to locate the appropriate version reference.

Alternatively, the process may know the version number of the particular version and, based on that version number, know which version reference to retrieve without having to scan each version reference sequentially. For example, the process may determine that the version number is '3' and, knowing that each version reference is N bytes, reads the version reference at $(3-1)*N$ bytes from the beginning of the set of version references (where 0 bytes from the beginning of the set of version references is the start of the value for the first version reference).

Vector Queries

[0121] A "vector query" is a query that targets one or more vectors in a vector database, such as vector database **120**. Vector database server **110** receives a vector query, generates an execution plan for the vector query, and processes the execution plan in order to retrieve one or more vectors from vector database **120**. A vector query typically includes a "query vector" and, optionally, one or more other search criteria. A query vector is a vector that vector database server **110** uses to identify one or more vectors in vector database **120**. Examples of one or more other search criteria include dates, numbers, strings, etc. For example, a vector query may ask for the top five matching vectors that are associated with the state of California and a date range between Feb. 1, 2024 and Mar. 5, 2024. Such other search criteria may comprise data from columns that are part of the same table that includes the vector column that stores the vectors that the vector query targets.

[0122] In order to identify vectors that are similar to a query vector, vector database server **110** may compare the query vector to each vector stored in vector database **120**. However, comparing a query vector to each vector in vector database **120** may take a significant amount of time that users are not willing to wait in order to receive an answer. Also, performing such a naïve scan of vector database **120** given a query vector may require a significant amount of computer resources that could be used for other tasks. To address these problems, a vector index may be generated and used in query vector processing.

Vector Indexes

[0123] Currently, similarity searches are often performed on data sets with billions of vectors (i.e., vector embeddings). For example, the Deep1B dataset contains 1 billion images generated by a

Convolutional Neural Network (CNN). Computing VECTOR_DISTANCE with every vector in the corpus to find Top-K matches at 100% accuracy is very slow. As a result, approximate vector indexes are used to trade-off search quality (recall/accuracy) for search speed.

[0124] Vector indexes tend to group data based on vector similarity with the search restricted to a few groups, achieving significant data pruning. Vector similarity is defined in terms of `vector_distance()` calculations. A goal is for a vector index to fit in memory (as opposed to only in slower, long-term (e.g., non-volatile) storage, such as disk) to allow for fast traversals and scans. With modern techniques and memory capacities, a index for billions of vectors can fit into volatile memory. Memory in modern computing machines is large enough for all but the largest workloads of giant web companies.

[0125] Two types of vector indexes that may be used to index vectors include Hierarchical Navigable Small Worlds Index (HNSW) and Inverted File Index (IVF). HNSW is an in-memory graph that is fast and relatively accurate, but it is larger in size than IVF. IVF is slower and less accurate, but it is smaller in size. Product Quantization (PQ) is a lossy compression technique that may be used to reduce the size of a vector index so that the index may fit into memory or be scanned faster. However, a tradeoff of PQ is lower accuracy. HNSW and IVF may be combined (with or without PQ) to optimize both speed and size.

Vector Index: IVF

[0126] An IVF index is based on K-means clustering or partitioning. A K-means clustering algorithm is applied to a set of vectors to generate K partitions. The value of K (or the number of partitions) may be based on the number of vectors. For example, $K = \sqrt{N}$, where N is the number of vectors. Each partition is identified by a centroid, which is a value that is conceptually the average of the vectors that are assigned to that partition. The centroid of a partition may be considered the “center of gravity” of the partition.¹ A goal in determining a centroid is to minimize a total distance between vectors within a partition and their centroid, so that each centroid is a good representative value for its partition.

[0127] FIG. 6 is a diagram that depicts an example set of five clusters **600** that is generated based on a clustering algorithm, in an embodiment. In this example, vectors are two-dimensional and, therefore, may be mapped onto a two-dimensional plane with an X-axis and a Y-axis. However, some vectors may have hundreds of dimensions. FIG. 6 also depicts a query vector **602** that is not in any of the five clusters **600**.

[0128] An IVF index comprises two types of tables: (1) a centroid table that stores all the centroids of all the partitions; and (2) K partition tables, each of which stores the vectors that are assigned to that corresponding partition based on closeness to the centroid represented by the partition. In a similarity search, given query vector **602**, the centroid table is searched first (referred to as the “first-level search”) to identify one or more centroids that are the most similar to (or has the lowest distance to) query vector **602**. Thus, either only a single centroid is selected from the centroid table or multiple centroids are selected from the centroid table. The number of centroids to select may be a default value (e.g., two) and/or may be based on vector distance from the query vector. (For example, select the closest three centroids that are within D vector distance from the query vector.) In the example of FIG. 6, a vector distance calculation is performed for each pair of vectors, each pair comprising query vector **602** and a different centroid of clusters **600**. Because there are five clusters **600**, five pairs are considered and five distance calculations are performed.

[0129] FIG. 7 is a diagram that depicts a subset of clusters that are selected based on distance calculations between query vector **602** and centroids in clusters **600**. In this example, two clusters are selected, which may be due to a threshold distance that each distance between a query vector and a corresponding centroid must be under in order to be considered a candidate cluster to search. Additionally or alternatively, two clusters are selected due to a pre-defined number of centroids to select (“n-probe”). In this example, the centroids in clusters #1 and #3 are selected from among the five centroids in clusters **600**. However, because cluster #4 is not part of the second-level search,

vector **702** cannot be considered a candidate vector. Also, vector **704** can be selected as a candidate vector, even though vector **702** is closer to query vector **602** than vector **704**.

[0130] Then, for each selected centroid, the partition to which that identified centroid belongs is searched to identify one or more vectors. This search is referred to as a “second-level search.” If the query vector is for the Top K, then the Top K vectors in each identified partition are identified and then the Top K from the Top K of each identified partition are selected. Even if the query vector is closer to a centroid associated with one partition, the closest vector to the query vector may be in another partition. Thus, searching one partition might not be sufficient for an accurate search; searching multiple partitions is generally prudent.

[0131] However, in an embodiment, the closer that a query vector is to the closest centroid, the fewer partitions are considered in the second-level search. A measurement of closeness of a query vector to the closest centroid may be based on one or more distances between the query vector and one or more vectors in the partition of the closest centroid and/or one or more other centroids. For example, if a query vector is over three times closer to the closest centroid than to a particular centroid, then the partition that corresponds to the particular centroid is not searched in the second-level search, neither any partition whose centroid is farther from the query vector than the particular centroid. As another example, if a query vector is closer to the closest centroid than the query vector is to over 50% of the vectors in the partition that corresponds to that centroid, then no other partition is searched.

Vector Index: Hnsw

[0132] HNSW is a multi-layer in-memory graph index that has relatively high speed and accuracy relative to IVF and other vector indexes. The graph index comprises vertices, each vertex corresponding to a vector. The lowest layer of the graph contains vertices of all of the vectors in the indexed data set. Higher layers of the HNSW index have a decaying fraction of the vertices in the layer below. In each layer, vertices are connected to their approximate M closest neighbors using edges that are used to walk the graph. At the lowest layer (“layer 0”), the number of neighbors of each vertex may be different than the number of neighbors of each vertex at higher layers, such as 2 M. The vertices at higher layers are on average much farther from each other (relative to lower layers) and, therefore, allow traversal of long distances.

[0133] Two major parameters for HNSW index construction are M and R. “M” is referred to as the “neighbor count” and is the number of neighbors that each vertex is connected to on each layer. Layer 0 (the lowest layer) may have double that number (e.g., 2 M neighbors). A probability distribution function may be defined based on M in order to determine whether a vertex is to be inserted in a layer that is above the lowest layer. The probability distribution function is such that probabilities decay with higher layers. When the probability drops below 1e-9, then no more layers are added. An example probability distribution function is

$$[00002] p(\text{Layer}) = (1 - 1 / M) / M^{\text{layer}}$$

[0134] Regarding parameter R, when a new vertex (corresponding to a new vector) is inserted into an HNSW index, a random number R between [0.0, 1.0] is generated. A new vertex is always inserted into layer 0. A new vertex is inserted into higher layers up to layer “i” if:

$$[00003] R > p(i - 1) + p(i - 2) \cdot \text{Math.} + p(0)$$

In an example, with M=10, if R=0.991, then the new vertex is inserted in layers 0 and 1.

[0135] An HNSW parameter that is used only for construction is referred to as “efConstruction” and refers to the number of vertices to consider within a layer when looking for the closest M vertices (or closest 2 M vertices in layer 0) to which to connect a new vertex. Larger values for this parameter improve index quality but slow down construction. An example value for this parameter is 2 M (or 4 M for layer 0).

[0136] An HNSW parameter that is used in searching is referred to as “efSearch” and refers to the number of vertices to remember in each layer when searching for the K nearest neighbors in a top-K search. Larger values of this parameter improve search quality but slow down searches. An

example value for this parameter is $2 \times K$ (or double the number of desired K matches). [0137] FIG. 8 is a diagram that depicts an example logical representation of an HNSW index 800 that comprises four layers, in an embodiment. An index search begins from an entry vertex 810 in the top layer (layer 3) and traverses edges looking for the vertex, in that layer, whose vector is nearest query vector 802. If M is 32, then 33 distance calculations are performed between query vector 802 and (i) the vertex of entry vector 810 and (ii) 32 neighbors of entry vector 810. The search process does not only consider neighbors of the entry vector 810, the search process finds the closest neighbor and computes distances for its neighbors as well. This process continues until all the neighbors of a vertex are further from the query vector than the vertex that was used to arrive in that neighborhood. Once the closest vector to query vector 802 in a layer is found (i.e., vertex 812 in this example), the search continues starting with that vertex in the next layer down. This process repeats for each intermediate layer of HNSW index 800. (Thus, vertex 822 is selected in layer 2 and vertex 832 is selected in layer 1.) The index search completes in the lowest layer (i.e., layer 0) by considering the (e.g., $2M$) neighbors of vertex 832, in the lowest layer, whose vectors are closest to query vector 802.

[0138] For traversing the lowest layer, two heaps are maintained: a candidates heap and a Top K heap. The candidates heap stores vertices that are candidates for further exploration and are ordered based on distance to the query vector. The Top K heap stores the current best Top K result set so far. The Top K heap holds efSearch vertices. The search proceeds as follows. Vertex 832 is at the top of the candidates heap and all its $2M$ neighbors are below it in the candidates heap, ordered by distance to query vector 802. Vertex 832 is selected and compared against all vertices in the Top K heap (which is initially empty). The goal is to check whether a vertex popped from the candidates heap can beat the worst vertex (in terms of distance to the query vector) from the Top K heap (which is also a priority queue ordered in reverse, i.e. furthest vertex among the efSearch vertices is at the top of the Top K heap). If the best candidate vertex beats the worst vertex from the current Top K heap, then the best candidate vertex is added to the Top K heap and the worst vertex is removed from the Top K heap. The search continues, meaning more candidates are explored. When vertex 832 is selected, the Top K heap is empty, so vertex 832 is automatically added to the Top K heap.

[0139] In a scenario where the Top K heap is full (i.e., it contains efSearch vertices), when a vertex V is selected from the candidates heap, the worst vertex in the Top K heap is replaced by V if V beats that worst vertex. All neighbors of V are then added to candidates heap and ordered within the candidates heap, and the next best vertex in the candidates heap is selected and the search continues. The search terminates if the current best candidate in the candidates heap cannot beat the worst vertex in the Top K heap. Increasing efSearch gives us accuracy because it is more likely for a candidate vertex to be better than the worst vertex in a set of one hundred vertices in the Top K heap than for the candidate vertex to be better than the worst vertex in a set of ten vertices. Thus, with larger values of efSearch, the exploration goes on longer.

Vector Index: HNSW Sizing

[0140] HNSW is best to use (compared to IVF) if the HNSW index fits in the memory of one server or computing device. The size of an HNSW index is dominated by layer 0. (Higher layers decline rapidly.) Layer 0 includes a vertex for each indexed vector and two vector identifiers (referred to herein as “VIDs”) for each edge connecting two vertices. Each VID may comprise four bytes.

[0141] For N vectors of size S with M neighbors each, the index size may be computed as follows:
 [00004] $\text{IndexSize} = \text{TotalSizeofVectors} + \text{TotalSizeofEdgeMetadata} = N * (S + M * 8)$

[0142] The following table shows sample index sizes for an HNSW index that is based on one hundred million vectors.

TABLE-US-00006	# of Dimensions	# of Neighbors	Index Size (GB)
512	16	203	512
32	215	512	
64	239	1024	16
394	1024	32	406
1024	64	430	2048
16	775	2048	32
787	2048	64	811

Thus, for a server with two terabytes of memory, the maximum number of vectors the server can store are approximately one billion vectors (of 512 dimensions each, each vector with 32 neighbors), five hundred million vectors (of 1024 dimensions each, each vector with 32 neighbors), and two hundred and fifty million vectors (of 2048 dimensions each, each vector with 32 neighbors).

HNSW: Example Storage Architecture

[0143] FIG. 9 is a block diagram that depicts an example storage architecture 900 for storing an HNSW graph, in an embodiment. Storage architecture 900 includes memory storage 910 and disk storage 920. Memory storage 910 is part of vector database server 110 and may be on a single compute instance or may represent the memory of multiple compute instances. Similarly, disk storage 920 is part of vector database server 110 and may be part of a single compute instance or may represent persistent storage of multiple compute instances.

[0144] Memory storage 910 stores layers 0-2 of an HNSW index along with a vertex vector map 912 (which is different than the base table), rowid table 914, private journals 916, and shared journal cache 918. Vertex vector map 912 comprises multiple rows, each row containing a vector and each row being associated with an index value that uniquely identifies a position within vertex vector map 912. Rowid table 914 comprises multiple rows, each row containing a row identifier and each row being associated with an index value that uniquely identifies a position within rowid table 914. The index position is the vertex ID (VID).

[0145] Multiple private journals 916 indicate that there are multiple pending transactions that have not yet committed, each private journal corresponding to a different pending transaction. Private journals 916 store data about vector modifications that are initiated by DML (data manipulation language) statements, such as inserts, deletes, and updates. Each of private journals 916 exists as long as a vector modification has not yet committed. At commit, one or more changes initiated by a DML statement are stored in shared journal table 924 in disk storage 920 (and, optionally, in shared journal cache 918). Periodically, or in response to certain events occurring or one or more criteria being satisfied, the contents of shared journal table 924 are applied to the HNSW index and vertex vector map 912. Thus, shared journal table 924 allows changes to the HNSW index to be buffered and to be applied to the HNSW index at a later time. Shared journal table 924 may also be used to allow “as of” vector queries, which is a vector query that asks for the state of the vector data “as of” a particular point in time in the past, but after the build/creation time of the HNSW index.

[0146] Shared journal cache 918 (1) enables transactionally-consistent queries involving vector search, (2) allows for delayed maintenance of the HNSW index, and (3) provides persistence for unapplied changes to the HNSW index, which enables faster restart after compute instance crashes. Regarding (1), the HNSW index and the shared journal cache 918 (and private journals 916 for in-transaction queries) always contain the changes to the set of indexed vectors since the creation of the HNSW index. Regarding (2), the HNSW index is best maintained in large batches of changes and shared journal cache 918 acts as a buffer of changes, thus, giving flexibility in the timing of HNSW index maintenance operations. Shared journal cache 918 may also be used for journaling during online creation of the HNSW index. Regarding (3), the HNSW index may be periodically persisted to persistent (non-volatile) storage using a checkpoint mechanism. Also, because shared journal cache 918 is persisted to shared journal table 924, changes are guaranteed to be persisted from the HNSW checkpoint SCN to the latest committed vector changes. Thus, upon restarting a compute instance after a crash, the HNSW index may be restored to memory from the checkpointed image, and the changes from shared journal table 924 may be applied to the restored image in a lazy manner.

[0147] Disk storage 920 also stores ROWID-VID table 922 which is a mapping table that maps rowids to vector IDs. The mapping table may be used to identify vectors that are deleted or those vectors that pass an attribute filter, described in more detail herein. One reason for using VID's instead of rowids in a vector index is because VID's are much smaller in size than rowids.

HNSW: Example Data Implementation

[0148] FIG. **10** is a block diagram that depicts an example organization of an example HNSW index **1000**, in an embodiment. In this example, HNSW index **1000** comprises three layers, layers 0-2. Each layer stores data about a set of vertices and contains a neighbor count array for the set of vertices and a neighbors array for the set of vertices. Thus, layer 2 includes a neighbor count array **1022** and a neighbors array **1024**. Each entry in neighbor count array **1022** corresponds to a different vector that has been assigned to layer 2. Each position of neighbor count array **1022** is an index value that points to a position within vectors array **1030**, which stores vector embeddings for the vertices in layer 0. Thus, the first position in neighbor count array **1022** corresponds to the vector in the first position of vectors array **1030**, the second position in neighbor count array **1022** corresponds to the vector in the second position of vectors array **1030**, and so forth. The same is true for neighbor count array **1012** and neighbor count array **1002**. As FIG. **10** indicates, the number of vertices assigned to each layer increases from the top-most layer to the bottom-most layer. Also, a vertex that appears in a top layer (e.g., vertex with Vector ID 1) or in an intermediate layer also appears in each layer below that layer.

[0149] In an embodiment, vectors array **1030** is constructed by assigning VIDs to vertexes during their respective insertion into a layer of HNSW index **1000**. Thus, the vertices (and, thus, their corresponding vectors) that are assigned to the highest layer will have lower VID/position numbers (and will be inserted first into HNSW index **1000**) relative to vertices that are assigned first to a lower layer.

[0150] Each entry in neighbor count array **1022** includes a value that indicates a number of neighbors that the corresponding vertex has been assigned during the index construction phase. Each (“referencing”) entry in neighbor count array **1022** also references or points to an (“referenced”) entry in neighbors array **1024** where at least one neighbor of the corresponding vertex is identified. That referenced entry may be the first entry of a set of entries that correspond to the neighbors of the referencing entry. Because each vertex has at most M neighbors, the remaining neighbors of the corresponding vertex are listed in entries subsequent to that first referenced entry, up to M-1 entries away from the first entry. As the example of FIG. **10** indicates, some vertices may have less than M neighbors.

[0151] Neighbors array **1024** includes vector IDs of neighbors of vertices indicated in neighbor count array **1022**. While neighbors array **1024** is presented as a single array, neighbors array **1024** may be implemented as a single array or as multiple smaller arrays (e.g., of size M) but are concatenated together for ease of description.

HNSW: Updates to HNSW Index

[0152] After an HNSW index is built, one or more changes may be made to the underlying vector data. Such changes may be initiated by DML statements. As described herein, one way to process the one or more changes is to buffer those changes by storing the changes in a separate table, such as shared journal table **924**. Eventually, it becomes necessary to apply those changes to the HNSW index. Another way to process the one or more changes is to apply the changes immediately to the HNSW index, without buffering. However, immediately applying changes made by a transaction means that the transaction “pays” the overhead of maintenance, thus slowing down the transaction. Regardless of which approach is used to process changes, whether in batches or incrementally as the changes are received, it is preferred to not rebuild the HNSW index “from scratch,” since that is an expensive operation in terms of time and resources.

[0153] FIG. **11** is a block diagram that depicts an updated version of HNSW **1000** after multiple inserts have been made into that index, in an embodiment. HNSW **1100** is the updated version of HNSW **1000** and includes the same data structures as depicted in FIG. **10** (except with updated reference numerals) as well as additional data structures. Layer 2 (the top-most layer) has layer-to-layer map **1126** and local VID to global VID map **1128**, while layer 1 has layer-to-layer map **1116**. In response to a batch update or an incremental update, it may be automatically determined (e.g.,

by an index updating component) that a vertex for the new vector is to be inserted into layer 2. A naïve approach would be to assign that new vertex to the next available VID in vectors array **1130** (e.g., 50001) and to use that VID in the other layers of HNSW **1100**. However, neighbor count array **1122** has a limited number of positions and it would be a waste of space to assign the new vertex to have a VID of 50000+, because then neighbor count array **1122** would have thousands of unused entries.

[0154] In the depicted embodiment, a new vertex/vector is assigned multiple local VIDs, each for a different layer of HNSW index **1100**. Thus, the next available VID in neighbor count array **1122** is determined. In this example, that next available VID is 5. However, VID 5 is already assigned to another vector in vectors array **1130**. Nevertheless, the new vector is also assigned the next available VID in vectors array **1130**, which is, in this example, 54252. Therefore, the new vector has two VIDs: 5 and 54252. In response, local VID to global VID map **1128** is updated to associate the “local” VID of 5 (local to layer 2) to the “global” VID of 54252.

[0155] Because this new vertex is assigned to layer 2, a vertex of the new vector is also assigned to layer 1. Similarly, because neighbor count array **1112** has a limited number of positions and it would be a waste of space to assign the new vertex to have a VID of 54252 in layer 1, the next available VID in neighbor count array **1112** is determined. In this example, that next available VID in layer 1 is 78. Although not depicted in FIG. **11**, layer 1 also has a local VID to global VID map, as in layer 2, so that vector distances may be computed during a search. Further, in this example, the new vertex in layer 1 has a neighbor (in layer 1) that is assigned VID 83 and, according to layer-to-layer map **1116**, is mapped to VID 55600 in neighbor count array **1102**. Thus, presuming that HNSW index **1000** had fifty thousand vectors when it was built, the vector assigned VID 55600 was added to the HNSW index sometime after HNSW index **1000** was built.

Transactionally-Consistent Hnsw Index

[0156] While the HNSW index is one of the fastest vector indexes, there is lack of support for transactional semantics. Also, current implementations of the HNSW index do not provide multi-version concurrency control, such as for large enterprises (e.g., financial institutions) who want to use AI Vector Search for real-time actions, such as fraud detection and anomaly detection.

Therefore, it is important to add transactional support to HNSW indexing strategies.

[0157] In an embodiment, an HNSW index supports transactional semantics. To support transactional semantics for HNSW indexes, four aspects are described: (1) efficient tracking of transaction changes; (2) efficient retrieval of transaction changes; (3) query results with consistent read (CR) semantics; and (4) accelerating Top K queries on a shared journal using local HNSW indexes.

[0158] Regarding (1) a DML transaction (i.e., the transaction that inserts one or more vectors, deletes one or more vectors, or updates/modifies one or more vectors) logs its changes in a transaction-specific in-memory private journal (e.g., one of private journals **916**), which is only visible to the transaction and is, therefore, not visible to other transactions. For new or updated vectors, the new/updated vectors are stored in addition to row identifiers (referred to herein as “rowids”) (identifying the rows in a base table where the new/updated vectors are stored) in separate in-memory extents.

[0159] When a DML transaction commits, its changes are flushed to an on-disk shared journal (e.g., shared journal table **924**) as well as optionally stored in an in-memory shared journal (e.g., shared journal cache **918**). In an embodiment, a shared journal comprises two tables with the transaction ID acting as the join key: (a) a CHANGE_LOG table that tracks the changes made by a transaction; and (b) a COMMIT_LOG table that tracks the commit SCN of every transaction.

[0160] For example, the CHANGE_LOG table may have at least three columns: one column for storing the transaction ID, another column for storing the type of modification (e.g., insert, delete, update), and another column for storing the inserted or modified vector. If the modification is a delete, then the third column may be blank or contain a null value. A single transaction may cause

multiple modifications to multiple vectors. Therefore, the same transaction ID may be repeated multiple times in the CHANGE_LOG table. Another column in the CHANGE_LOG table may be a transaction change number (XCN) that orders changes within a transaction. Also, the transaction ID column may be implemented as three columns: one column for the undo segment number (which corresponding segment is where the transaction generates undo), another column for the slot number (which is the slot, in an undo segment transaction table, that this transaction occupies), and another column for a sequence number (which is a counter that is increased on every reuse because slots can be reused). Additional columns in the CHANGE_LOG table may be for a DML opcode and a Vector payload.

[0161] The COMMIT_LOG table may have two or more columns: one column for storing the transaction ID and another column for storing a commit time (e.g., SCN) of the corresponding transaction, where the transaction ID column may be implemented as three columns. In this way, only a single commit time needs to be stored per transaction, as opposed to storing a commit time for each change that resulted from the transaction. A row, in the COMMIT_LOG table, that represents a transaction, is atomically patched in a data block to reflect the commit SCN. This step is important to ensure transactional visibility of changes in the CHANGE_LOG table.

[0162] In an embodiment, the COMMIT_LOG table also has a partition ID column that is used to partition the COMMIT_LOG table. For example, a default rule may be that every partition of the COMMIT_LOG table is bounded to 1 GB in size. The initial partition may have a partition_id of 1. Once that partition reaches 1 GB in size, the “current partition id” is changed to 2, and all transactions committing after that point will be associated with this value. In this way, transactions are grouped somewhat in timestamp (e.g., SCN) ranges.

[0163] Regarding (2) or the efficient retrieval of transaction changes, a common operation on the shared journal table may be to find the set of committed changes in a time window (e.g., an SCN window, such as between the SCN of the vector query/scan and the HNSW's build SCN). The shared journal is organized in a manner that makes this operation efficient. The shared journal may be list partitioned by a system-maintained incrementing partition_id column. This list partitioning ensures that there is no skew between partitions. New transactions are added to the latest partition. For example, a partition corresponding to partition_id=1 captures all transactions that committed between SCNs **101** and **200**. A partition corresponding to Partition_id=2 captures all transactions between SCNs **201** to **300**. When querying the shared journal, the query is by SCN range.

However, partition pruning cannot be taken advantage of because the shared journal is partitioned by partition_id. For example, a query is looking for all changes between SCN **150** to **180**. In order to prune the partition corresponding to partition_id=2, additional data structures are needed. This is where a partition-local B-tree index may come in. If a local B-tree index on COMMIT_SCN exists for each partition, then the minimum and the maximum values (the leftmost and rightmost leaf nodes) of the per-partition B-tree may be used to quickly prune out the partitions that are not interesting. Such an index is not built for a partition that is “open” (i.e. it is the current partition to which changes may be added) because it adds overhead to maintain the B-tree index. Instead, the system waits for the partition to be closed (i.e., the current partition_id advances) before creating an index for the closed partition.

[0164] FIG. **12** is a flow diagram that depicts an example process **1200** for recording changes to vectors and leveraging those changes during processing of a vector query, in an embodiment.

[0166] At block **1210**, a change request is received. The change request may be in the form of a DML transaction. The change request may identify a set of one or more vectors (whether to insert or delete) and/or may specify one or more criteria that a vector (and/or an entity associated with the vector) must satisfy in order to modify the vector.

[0167] At block **1220**, one or more changes to one or more vectors are stored in a private journal that is only visible to the transaction initiated by the change request.

[0168] At block **1230**, once the transaction commits, the one or more vector changes are stored in a shared journal, each change indicating a type of vector change and a commit time of the vector change. The shared journal is visible (i.e., accessible) (and, therefore, its contents) to a vector query processing engine. A single transaction may result in many vector changes and, therefore, multiple vector changes logically have the same commit time in the shared journal (because a single commit time is stored per transaction in the commit log table while the changes in the change log table reference this commit time by the transaction ID foreign key).

[0169] At block **1240**, a vector query is received. The vector query may have been composed by a user or a device that is remote relative to VDBMS **100**. The vector query may be associated with a timestamp or SCN. If so, then the vector query may specify (or otherwise indicate) the timestamp or the timestamp may be inferred based on when VDBMS **100** receives the vector query. If the vector query specifies the timestamp, then the timestamp may indicate a time that is before the current timestamp (or the receipt timestamp of the vector query) and after the timestamp (or SCN) of the build of the HNSW index.

[0170] The vector query may be processed in a transaction that is separate from the transaction in which the change request was processed. For example, the change request and the vector query may have been received, by VDBMS **100**, from different users or client devices, or by the same user/client device at different times. Alternatively, the vector query may have been processed in the same transaction as the change request. For example, the change request and the vector query may have been included in the same request, received by VDBMS **100**, from a particular user or client device.

[0171] At block **1250**, one or more changes in the shared journal are identified. Such identification may be based on a timestamp of the vector query, if such a timestamp exists. For example, if a change in the shared journal has a timestamp that is after the vector query timestamp, then the change is ignored. Conversely, if a change in the shared journal has a timestamp that is before the vector query timestamp, then the change is identified. A change of a vector may be a deletion of the vector, an insertion of the vector, or an update of the vector.

[0172] At block **1260**, a final result is generated based on a result set from traversing (scanning) the HNSW index based on the vector query and the one or more changes in the shared journal. If a change is deletion, then it is determined whether the VID of the change is in the result set. If so, then the VID is removed from the result set. This is a post-filter approach. Alternatively, in a pre-filter approach where deleted VIDs are identified first, deleted VIDs are not returned from the HNSW graph search if, during exploration of the HNSW graph, a deleted VID is never moved from the candidates heap to the Top K heap. If the change is an insertion of a vector, then an operation is performed on the vector, such as a vector distance calculation or similarity score indicating a similarity between the inserted vector and the query vector. If the vector distance calculation is less than the largest vector distance calculation in the result set, then the VID of the inserted vector is added to the result set and the VID Of the largest vector distance calculation is removed from the result set. If a change is an update, then the processes described above for delete changes and insert changes are followed.

[0173] In a related embodiment, block **1250** occurs before the HNSW index is traversed or scanned based on the vector query. In this way, the one or more changes are known before (and can be applied during) traversal of the HNSW index. Otherwise, traversal of the HNSW index followed by applying the one or more changes may result in too few vectors. For example, if a vector query is a Top K query, K results are identified as a result of traversal of the HNSW index, and one or more deletes are identified in the shared journal where the VIDs of those deletes are in the K results, then fewer than K results will be returned. Otherwise, the HNSW index may have to be traversed again in order to increase the final results to K in number.

Transactionally-Consistent HNSW Index: Consistent Read

[0174] Regarding (3) or query results with consistent read semantics, the private journal and/or

shared journal are used to guarantee consistent read (CR) semantics for Top K in-transaction and out-of-transaction vector queries. An in-transaction vector query is one that queries a set of data in the same transaction that caused the modification of the set of data. Thus, the vector query targets a set of data that was just changed by a DML transaction, but before the DML transaction commits, which would make the changes visible/accessible to other transactions and queries.

[0175] FIG. 13 is a flow diagram that depicts an example process 1300 for processing an in-transaction read associated with a vector query, in an embodiment.

[0176] At block 1310, a first set of rowids that have recorded deletes in a private journal is identified. The private journal may be implemented as a table, a linked list, an array, etc. Each entry in the private journal associates a type of change with a rowid, identifying a row in a base table that stores vectors. Thus, block 1310 may involve identifying each entry in the private journal that indicates a deletion and, if so, retrieving the associated rowid. Some iterations of block 1310 may result in identifying no rowid if no deletes were recorded in the private journal.

[0177] At block 1320, a first set of vector IDs corresponding to the first set of rowids is identified from the ROWID-VID table, described previously. For example, for each rowid in the first set, an entry in the ROWID-VID table is identified that has a matching rowid. The VID in that entry is then retrieved. Block 1320 may be skipped if the first set of rowids is empty.

[0178] At block 1330, a second set of rowids that have recorded deletes in the shared journal between the query timestamp (e.g., SCN) and HNSW build timestamp (or SCN) is identified. Block 1330 is similar to block 1310 except for the shared journal and the timestamp check. Thus, some iterations of block 1330 may result in identifying no rowid if no deletes were recorded in the shared journal.

[0179] At block 1340, a second set of vector IDs corresponding to the second set of rowids is identified from the ROWID-VID table. Block 1340 is similar to block 1320 except for a different set of rowids. Block 1340 may be skipped if the second set of rowids is empty.

[0180] At block 1350, a delete filter combining both sets of vector IDs (representing deleted vectors) is generated. The delete filter may be implemented as an array, a bitmap, or a list. The contents of the delete filter may be ordered or unordered. If ordered, the delete filter may be ordered by VID in an increasing or decreasing manner. The delete filter may be a bitmap such that each entry in the bitmap is associated with an index value that is based on VID. Thus, a VID may be mapped to a specific entry in the VID and if the entry includes one value (such as a bit value of '1'), then that indicates that the corresponding vector is deleted; otherwise the corresponding vector is not deleted.

[0181] At block 1360, a traversal of an HNSW index for a Top K vector query is performed. When a VID is about to be added to a result set, it is first determined whether the VID is in the delete filter and, if so, then that VID skipped and, therefore, is not added to a temporary result set. This approach gives one set of Top K results.

[0182] In a related embodiment, a vector query processing engine maintains two heaps: a candidates heap and a Top K heap. Both heaps are initially empty when traversal of the HNSW index begins. The candidates heap is a staging area before VIDs are added to the Top K heap, which contains the Top K results thus far. The Top K heap may be an ordered list of VIDs, ordered based on vector distance to the query vector. When a VID and its neighbors are identified while traversing layer 0 of the HNSW index, that VID and the VIDs of its neighbors are stored in the candidates heap. Then, for each VID indicated in the candidates heap, it is determined whether the delete filter contains that VID. If so, that VID is ignored. (The HNSW index remains unchanged, despite the deleted vector.) If not, then it is determined whether the VID is better (e.g., closer in vector distance to the query vector) than any of the VIDs in the Top K heap. If so, that VID from the candidates heap replaces, in the Top K heap, the VID of the "worst" vector (e.g., farthest from the query vector in terms of vector distance). The addition of a VID from the candidates heap to the Top K heap may retain the ordering of the Top K heap by vector distance.

[0183] At block **1370**, a Top K is computed for the private journal. Block **1370** may involve identifying all vectors that were inserted in the private journal, computing a vector distance of each identified vector to the query vector, and generating a Top K based on the computed vector distances. If K is equal to or larger than the number of inserts in the private journal, then all the corresponding vectors are returned as the Top K. Nevertheless, the vector distance is still computed for each identified vector.

[0184] Block **1370** may involve determining whether any vectors that are identified as inserted in the private journal were subsequently deleted. Thus, block **1370** may involve determining, for each inserted vector, whether that inserted vector is indicated in the private journal as deleted (and that deletion has a timestamp within the timestamp window of the vector query). Therefore, vectors that were inserted and then subsequently deleted (according to their respective timestamps or to their order in the private journal) are ignored when computing the Top K for the private journal.

[0185] At block **1380**, a Top K is computed for the shared journal. Block **1380** is similar to block **1370**, except with respect to the shared journal instead of the private journal and considering deletes from the private journal. Thus, block **1380** may involve identifying all vectors that were inserted in the shared journal, excluding vectors associated with rowids that are identified as deleted in the private journal, computing a vector distance of each identified vector to the query vector, and generating a Top K based on the computed vector distances.

[0186] Block **1380** may involve determining whether any vectors that are identified as inserted in the shared journal were subsequently deleted. Thus, block **1380** may involve determining, for each inserted vector, whether that inserted vector is indicated in the shared journal or the private journal as later deleted (and that deletion has a timestamp within the timestamp window of the vector query). Therefore, vectors that were inserted and then subsequently deleted (e.g., according to their respective timestamps or to their order in the shared journal) are ignored when computing the Top K for the shared journal.

[0187] Blocks **1360-1380** may be performed in parallel and, together, produce three sets of Top K results.

[0188] At block **1390**, the Top K results of each of blocks **1360-1380** are combined to generate a final or “true” Top K results. The final Top K may comprise only vectors from block **1360**, only vectors from block **1370**, only vectors from block **1380**, or any combination of vectors from blocks **1360-1380**. In a related embodiment, block **1390** may involve first merging/combining the Top K results from block **1370** with the Top K results from block **1380**. The merged results are then merged/combined with the Top K results from block **1360**. Since the HNSW index is likely to be much larger than even the combined sizes of the private journal and the shared journal, the Top K results from blocks **1370** and **1380** might be available much sooner than the Top K results from block **1360**. However, even though the HNSW graph contains more vectors than the private and shared journals, the search of the HNSW graph only scans $O(\log N)$ of vertices. Depending on the rate of DMLs and how frequently the HNSW graph is refreshed, the number of vectors in the shared journal may exceed $O(\log N)$ and, therefore, the Top K results from block **1360** might be available before the Top K results from block **1380**.

[0189] In a related embodiment, one or more of blocks **1370** and **1380** are intentionally not performed. This may allow vector query processing to complete sooner and, optionally, achieve one or more time/resource constraints. For example, a vector query may be associated with one or more of a time constraint or a resource constraint. If performing either block **1370** or block **1380** or both would cause the vector query to exceed the time constraint (e.g., five seconds), then that block is not performed. Similarly, if performing either block **1370** or block **1380** or both would cause the vector query to exceed the resource constraint (e.g., greater than 100 MB of memory for vector query processing), then that block is not performed. As another example, a user or entity that initiates the vector query may be associated with an accuracy expectation, such as a threshold, rating, or category. (The association may be an explicit association that was specified in the vector

query or may be a default association that was established previous to receiving the vector query.) If satisfying the vector query without using the private or shared journals satisfies the accuracy expectation of the vector query, then scanning the private and shared journals for inserts may be skipped. However, scanning the private and shared journals for deletes and constructing a delete filter may still be performed.

[0190] In a related embodiment to the embodiment of FIG. 13, a similar process is performed for an out-of-transaction read. One difference is that there are no blocks (or steps) for reading data from a private journal and, therefore, no Top K result generation based on a private journal.

Transactionally-Consistent HNSW Index: Partition-Specific HNSW Indexes

[0191] Regarding (4) or accelerating Top K queries, one approach for processing a Top K vector query is to search the entirety of the shared journal for modifications to vectors that are reflected in the HNSW index. In an embodiment, an HNSW index is built on each partition of the shared journal (or CHANGE_LOG table), except for the latest partition, which may still be continually modified by DML transactions. Thus, a partition-specific HNSW index may be built for a particular partition immediately after it is determined that no more transactions may be added to the particular partition.

[0192] In a related embodiment, a partition-specific HNSW index is only built for partitions that satisfy one or more criteria. Example criteria include a partition being greater than a certain size (e.g., in bytes or number of rows) and a partition having a minimum number of inserts.

[0193] Once partition-specific HNSW indexes are built on top of one or more shared journal partitions, processing Top K vector queries involves accessing those partition-specific HNSW indexes. If there are multiple partition-specific HNSW indexes, then they may be accessed in parallel during query processing, which further improves vector query response times. For partitions that do not have an HNSW index built for them, those partitions will need to be scanned directly (e.g., one entry or row at a time) without the benefit of a vector index.

[0194] In a related embodiment, a vector query includes a time range, such as a SCN range. A vector query execution engine may use this vector query time range to identify one or more partitions whose time ranges (or SCN ranges) do not overlap the query time range. Those one or more partitions (or any corresponding partition-specific HNSW indexes) are excluded from consideration (or not accessed) during vector query processing, as described previously.

Multi-Snapshot HNSW Index

[0195] As described herein, an HNSW index is a multi-layer in-memory neighbor graph built from vectors. The HNSW index is also built as of a specific point in time, which may be determined using timestamps or system change numbers (SCNs). The time at which an HNSW index is built is referred to as the “build SCN.” Thus, an HNSW index may be considered a “snapshot” of the index. As DML statements are applied to the base table of vectors upon which the HNSW index is built, this snapshot becomes stale, leading to slower query performance due to ever-increasing disk-based shared journal access. Therefore, the most recent snapshot eventually should be refreshed. Refreshing a snapshot refers to updating a snapshot based on changes that have been made to the base table of vectors since the build SCN of the snapshot.

[0196] There are two approaches for refreshing a snapshot: a tic-toc snapshot approach and a multi-snapshot approach. In the tic-toc snapshot approach, at most two snapshots of the HNSW index are maintained or tracked: the current active snapshot and a new snapshot that is built in the background that eventually becomes visible to vector queries once fully built. Once the new snapshot is available, vector queries that have a timestamp or SCN that is after (timewise) the new snapshot's build SCN are directed to the new snapshot. Vector queries that began to be processed against the old snapshot eventually end (whether terminated due to an error or fully processed with a query result generated). Once a retention threshold has elapsed, it is guaranteed that VDBMS 100 will not receive any vector queries with scan SCN that is less than the new snapshot's build SCN. The old snapshot may be deleted after the lapse of that retention threshold.

[0197] In the tic-toc snapshot approach, long running queries (or even large retention thresholds) can prevent reclamation (or deletion) of the old snapshot, which also prevents creation of a newer snapshot since only two snapshots are allowed. The tic-toc snapshot approach may be generalized to a multi-snapshot design where snapshots are periodically created and tracked.

Multi-Snapshot HNSW Index: Heuristics for Snapshot Creation Decision

[0198] A number of heuristics may be used to determine whether to create a new snapshot of an HNSW index. One example heuristic is the number of DML transactions recorded in the shared journal. If the number of DML transactions is greater than a certain threshold, then a snapshot creation process is initiated. However, not all DML transactions have the same impact on increasing vector query processing time. For example, delete changes have a lesser impact (i.e., deletes may be processed faster than inserts or updates). Therefore, some types of changes may be weighted differently than other types of changes. For example, a delete may be considered half of an insert and a threshold number of changes indicated in a shared journal is fifty changes. Thus, if there are thirty inserts and thirty deletes in the shared journal, then, though there are literally sixty changes, because a delete is considered half of an insert, the total number of changes associated with the shared journal is forty-five. Because forty-five is less than fifty, then another snapshot is not created.

[0199] Another example heuristic is an average rate of incoming DML transactions. If the average rate exceeds a pre-defined threshold, then a new snapshot is created. A weighted average may be computed over certain time windows (e.g., every ten seconds) to determine if the rate of incoming DML transaction has exceeded the threshold.

[0200] Another example heuristic is a number of scans of the HNSW index. If a query optimizer selects non-indexed access of vectors, then creation of a new snapshot may be delayed, even in the presence of relatively many DML transactions.

[0201] Another example heuristic is memory footprint/budget. Each snapshot occupies some memory of a compute instance. The memory that is used to store vector-related data (e.g., a base table of vectors and a vector index) is referred to as “vector memory area.” Depending on the amount of available memory, snapshots can be created more or less aggressively. For example, if the vector memory area is only 10% utilized and no previous snapshot is greater than 2% of the vector memory area, then new snapshots may be created relatively frequently. On the other hand, if the vector memory area is 90% utilized, then new snapshots are created less frequently and only when considered “necessary.” Also, older snapshots may be deleted more aggressively.

[0202] Another example heuristic is an average latency of vector queries against the HNSW index. For example, if the average latency of vector queries against the HNSW index in a most recent time window is 20% greater than an average latency immediately after the build time of the HNSW index, then another snapshot of the HNSW index is created.

Multi-Snapshot HNSW Index: Efficient Creation of Snapshots

[0203] In an embodiment, snapshots of an HNSW index are created incrementally. This means that snapshot $i+1$ is created by applying changes in the shared journal that is associated with the HNSW index and, thus, stores changes that have a timestamp that is after the build timestamp of snapshot i . In other words, the changes in the shared journal are associated with timestamps, or SCNs, in the SCN range (Snapshot $\#i$ creation SCN, Snapshot $\#i+1$ chosen creation SCN] and are applied to snapshot i . For example, a snapshot creation process (e.g., executing in vector database server **110**) identifies the build SCN of snapshot i , identifies a current SCN that will be the build SCN of snapshot $i+1$, identifies (in the shared journal) all changes associated with an SCN between those two SCNs, and applies those identified changes to snapshot i (or a copy thereof), resulting in snapshot $i+1$. Thus, creating a new snapshot from an old (or most recent) snapshot may involve copying the old snapshot (with all its vertices and neighbor lists) and modifying that copy.

[0204] There are two main types of changes that are to be accounted for when creating a new snapshot: deletes of vectors and inserts of vectors. (An update of a vector may be implemented as a

deletion of the vector followed by an insert of a new vector that replaces the vector.)

[0205] Deletes may be handled in one of multiple ways. In a first deletion technique, for each deletion of a vector, a vertex, in the new snapshot, is identified that corresponds to the deleted vector and that vertex is marked as deleted or masked off. However, the neighbor lists of that vertex remain unchanged. Then, during query execution, a Top K traversal of the new snapshot skips or avoids returning those marked vertices, which process is similar to the handling of the delete filter.

[0206] In a second deletion technique, neighbor lists (copied from the most recent snapshot) are adjusted to remove, in the new snapshot, the deleted vertices. Whenever a vertex V is removed, it affects the vertices that had V as one of their 2 M neighbors. The set of all such vertices comprises the proximal graph. Thus, a proximal graph comprises (1) a first set of vertices, in a snapshot, that are deleted based on changes indicated in the shared journal and (2) a second set of vertices, in the snapshot, each of which is connected to at least one vertex in the first set of vertices. For each vertex in the second set, an alternate neighbor is chosen to replace V. Alternatively, each vertex in the second set is “re-inserted” into the graph. The insertion algorithm takes care of adjusting neighbor lists.

[0207] In an embodiment, one or more factors are considered when determining which delete technique to follow. Example factors include the number of deletes and a percentage of all vectors that are deleted. The greater the number of deletes, the more likely that the second technique is implemented rather than the first technique. For example, if the number of deletes is lower than a threshold (e.g., whether a particular number of as a percentage of the total number of indexed vectors in the base table, such as 1%), then the first technique is implemented; otherwise, the second technique is implemented.

[0208] In some cases, the number of deletes is so great or the deleted vertices are in so many neighbor lists that it would be better to rebuild the HNSW index rather than to create a new snapshot from a previous snapshot. As deletes accumulate, search latency increases because, even though the graph can be explored through masked deleted vertices, those deleted vertices should not be returned, and hence it takes longer to find the Top K. If the size of the proximal graph is large (for example, due to an articulation point/cut vertex/hub vertex of the graph being deleted) or is greater than a threshold percentage of the most recent snapshot, then the entire HNSW index is rebuilt “from scratch” to create the new snapshot. (An articulation point/cut vertex/hub vertex are synonyms referring to a popular neighbor that, when deleted, can affect multiple nodes in a graph.) This is done to avoid reducing the quality of results because key vertices were deleted. Therefore, one or more factors, such as the number of deletes, the number of neighbor lists with deleted vertices, and the size of the proximal graph are considered when determining whether to rebuild the HNSW index instead of starting with a snapshot.

[0209] Regarding inserts, inserts are handled using the insertion process described herein. The number of inserts and/or the percentage of all vectors that are inserted may be factors in determining whether to rebuild the HNSW index instead of creating a new snapshot from a current snapshot.

[0210] In an embodiment, a copy-on-write technique is used to create new copies of neighbor lists that have been modified. For neighbor lists that have not been modified, the memory from the previous snapshot is shared in the new snapshot. This ensures that the memory overhead of the new snapshot is minimal.

[0211] As depicted in FIG. 10, each layer has a neighbor count array and a neighbors array. A neighbor count array is essentially a “source array.” Index 1 in the source array is vertex ID 1, index 2 in the source array is vertex ID 2 and so on. The neighbors array stores up to M (or 2 M) neighbors per source vertex. There is a pointer from every source array vertex to its neighbors (specifically, the first of its M neighbors).

[0212] With that context, using the copy-on-write technique when creating a new snapshot, a new

copy of the entire source array is created in each layer. Essentially, every source vertex “points” to the existing neighbor list (unchanged vertex) or to a new neighbor list (because the source vertex's neighbors have been affected). Because a new copy of the source array is created, each vertex points to its old neighbor lists by default. If the neighbors of a source vertex have changed, then a copy-on-write is performed for the neighbor list of that source vertex. This means that a copy of the M neighbors is created elsewhere, the old deleted neighbors are removed from that copy, new neighbors are added to the copy, and the pointer of the source vertex entry to this copy is updated. Thus, the set of unchanged neighbor lists are pointed to by the source array of snapshot 1 and snapshot 2. This neighbor list memory is shared. For the changed neighbor lists, the source array for snapshot 1 points to a different memory region than the source array for snapshot 2.

[0213] In a related embodiment, neighbor lists are broken up into chunks (e.g., of 1 MB), rather than a single contiguous entity of vertices. Within each chunk, if any neighbor list is changed, then a copy-on-write is performed on the entire chunk. Therefore, it is possible that even unchanged neighbor lists get copied just because they happen to be within the same chunk that holds the neighbor list of a vertex that needs to be changed. This embodiment allows memory to be freed at the chunk boundary when the old snapshot is reclaimed. A more advanced embodiment is to actually manage memory within the chunk as essentially a free list of neighbor lists, each entry is M in size. Thus, copy-on-write is performed only on the actual changed neighbor lists.

[0214] When a snapshot is reclaimed, that region of M neighbors within the chunk may be marked as free, and this memory can be used to perform a copy-on-write on the neighbor list of some other snapshot.

[0215] In summary, in response to determining to generate a new snapshot of an HNSW index, a first set of vertices, in the latest (or most recent) snapshot, whose neighbor lists have not changed since that snapshot is identified. Also, a second set of vertices, in the latest snapshot, whose neighbor lists have changed since that snapshot is identified. Memory, for the latest snapshot, that stores neighbor lists of the first set of vertices is shared with the new snapshot. For example, the upper two layers of the two snapshots may be identical and, therefore, completely shared when traversing/scanning either snapshot for different vector queries with different timestamps. Unused memory is allocated for the new snapshot in order to store the updated neighbor lists of the second set of vertices. (The source array is a new piece of memory for each snapshot.) This allocation may involve copying the memory, for the latest snapshot, that stored the neighbor lists of the second set of vertices into the unused memory for the new snapshot and then modifying the neighbor lists so that they no longer include deleted vertices.

Multi-Snapshot HNSW Index: Selecting a Snapshot for a Vector Query

[0216] In response to receiving a vector query (e.g., at VDBMS **100**), a snapshot is selected from among multiple snapshots of an HNSW index, if multiple snapshots exist. If only a single snapshot exists or is stored for the HNSW index, then only that snapshot is selected.

[0217] Selecting from among multiple snapshots involves comparing a timestamp (e.g., SCN) of the vector query with a timestamp of one or more snapshots. The most recent snapshot that has a build time that is less than the timestamp of the vector query is selected as the snapshot to scan or traverse. All snapshots are associated with the same shared journal, which contains changes to vectors in the vector table, where the changes were received after the build time of the original snapshot. While the shared journal is shared across snapshots, changes that occurred after any snapshot was created may be identified using timestamps (e.g., SCNs) of those changes. Thus, the shared journal may contain changes that have not yet been reflected in any snapshot. The process described herein regarding consistent read (CR) semantics may be used to obtain results of the vector query.

[0218] Executing the vector query against the selected snapshot and the shared journal involves traversing the particular snapshot based on a query vector that is included in the vector query to generate a first set of results. Also, a second set of results is generated by scanning the shared

journal based on the timestamp of the vector query. The shared journal may comprise (1) first changes that are associated with timestamps that are before the vector query's timestamp and (2) second changes that are associated with timestamps that are after the vector query's timestamp. Thus, executing the vector query against the shared journal involves executing the vector query against the first changes and not the second changes.

[0219] Instead of selecting the most recent snapshot whose build time is older than the timestamp of the vector query, it is possible to select older snapshots as long as the shared journal has the appropriate changes. However, a primary purpose of creating snapshots is to reduce the potentially unindexed Top K computations on the shared journal. Therefore, there is little utility in selecting an older snapshot for a vector query, unless the most recent snapshot is somehow unavailable.

Multi-Snapshot HNSW Index: Efficient Reclamation of Older Snapshots

[0220] In an embodiment, a snapshot is automatically reclaimed (deleted), freeing up its memory to be used for other purposes. A snapshot may be reclaimed under different scenarios. For example, if no queries are actively accessing a snapshot, then the snapshot may be reclaimed. Determining that no queries are actively accessing a snapshot may be through a reference counting mechanism, which may be executing on one or more compute instances in a cluster. For instance, the reference counting mechanism (RCM) maintains a counter for each snapshot of multiple snapshots. The RCM increments a counter for a snapshot for each vector query that VDBMS **100** receives and that is executed against the snapshot. The RCM decrements the counter when the snapshot is no longer accessed by a process executing the vector query or when the vector query completes executing.

[0221] As another example, if no queries are active in a vector database that could potentially access a snapshot, then the snapshot may be reclaimed. This can be guaranteed by waiting for a retention threshold (e.g., fifteen minutes by default) to elapse since the creation of the snapshot. An error may be raised indicating that a snapshot is too old if vector queries (with timestamps older than the retention threshold) target the HNSW index. Such a scenario might happen if a vector query was issued a while ago, but the vector query went into an I/O wait or executing the vector query involves evaluating extremely complex filters before the Top K operation is executed.

[0222] Flashback queries allow for vector queries that specify a timestamp that is older than the retention threshold. To allow such queries to use an HNSW index, significant amounts of memory are needed to keep older snapshots around. Alternatively, such queries can fallback to a table scan, which is slower than using a vector index, but a table scan provides correct results.

[0223] Another scenario in which a snapshot may be reclaimed is if the overall footprint of snapshots has exceeded an allotted memory budget for the HNSW index. The memory budget is for all snapshots of the HNSW index. If there are multiple HNSW indexes, then each HNSW index may have an index-specific memory budget.

[0224] For example, a snapshot creation process checks a current allotted memory budget for an HNSW index, determines a total size of the current snapshots of the HNSW index, estimates a size of a new snapshot, and determines whether the current allotted memory budget is greater than the sum of total size of the current snapshots and the estimated size of the new snapshot. If not, then a snapshot is reclaimed, such as the oldest snapshot, which is less likely to be targeted by a future vector query.

Multi-Snapshot HNSW Index: Efficient Reclamation of Shared Journal Contents

[0225] In an embodiment, contents of a shared journal are reclaimed if one or more conditions are satisfied. For example, contents of a shared journal are reclaimed if the in-memory snapshot that depends on those contents can no longer be used by vector queries or if the in-memory snapshot has already been reclaimed. For an HNSW index that is duplicated across a cluster of compute instances, this rule may apply to all copies of the HNSW index.

[0226] Another condition for reclaiming contents of a shared journal may be that a newer, in-memory snapshot that has a timestamp that is after the timestamp of the contents to be reclaimed has already been persisted. As described herein, a shared journal table comprises a CHANGE_LOG

table and a COMMIT_LOG table that are partitioned by a partition ID that is designed to keep the size of each partition the same. A partition may be thought of as tracking changes for transactions within a timestamp range of the changes in the partition. When a snapshot is persisted, each change that is associated with a timestamp that is before the build timestamp of the snapshot may be reclaimed.

[0227] In an embodiment, snapshots of an HNSW index are persisted (e.g., stored to disk) to ensure quick population after instance crash/shutdown and restart. However, such persistence may be optional if an HNSW index is rebuilt from scratch upon a restart.

Multi-Snapshot HNSW Index: Example Process

[0228] FIG. **14** is a flow diagram that depicts an example process **1400** for processing a vector query in the context of multiple snapshots of an HNSW index, in an embodiment. Process **1400** may be performed by one or more components of vector database server **110**.

[0229] At block **1410**, multiple snapshots of an HNSW index are stored. Block **1410** may involve storing the multiple snapshots on a single compute instance or on each of multiple compute instances. Alternatively, a different subset of snapshots of the multiple snapshots may be stored on a different compute instance of multiple compute instances. Each snapshot has a different build time.

[0230] At block **1420**, a vector query that is associated with a particular timestamp is received. The vector query may originate from a client device that is separate from VDBMS **100**. For example, the client device may be remote relative from VDBMS **100** and may transmit the vector query over a computer network, such as a LAN, WAN, or the Internet. The timestamp may be earlier time than the current time. The timestamp may be in the form of an SCN.

[0231] At block **1430**, a particular snapshot from among the multiple snapshots of the HNSW index is selected based on the particular timestamp. For example, the most recent snapshot that is prior to the particular timestamp is selected. The particular snapshot may be the latest snapshot, may be the first snapshot, or may be an intermediate snapshot that is “between” (in terms of time and evolution) the latest snapshot and the first snapshot.

[0232] At block **1440**, a shared journal of the HNSW index is identified. The shared journal stores a set of changes to vectors that are stored in the vector table.

[0233] At block **1450**, the vector query is executed against the particular snapshot and relevant changes stored in the shared journal. For example, a first set of results is generated by traversing the particular snapshot based on a query vector that is included in the vector query. Also, a second set of results is generated by scanning the shared journal based on the particular timestamp. The shared journal may comprise (1) first changes that are associated with timestamps that are before the particular timestamp and (2) second changes that are associated with timestamps that are after the particular timestamp. Thus, executing the vector query against the shared journal involves executing the vector query against the first changes and not the second changes. Block **1450** may involve combining the first set of results and the second set of results.

[0234] FIG. **15** is a flow diagram that depicts an example process **1500** for creating a new snapshot of an HNSW index, in an embodiment. Process **1500** may be performed by one or more components of vector database server **110**.

[0235] At block **1510**, it is determined whether a new snapshot is to be generated. One or more factors may be considered in making this determination, such as a number of changes in the shared journal, an average rate of incoming changes to the vector table, a number of scans of the HNSW index, a latency of one or more vector queries against the latest snapshot, and an amount of memory that is required to store all the snapshots. If the determination is positive, then process **1500** proceeds to block **1520**, otherwise, process **1500** ends. The process for reclaiming snapshots and contents of the shared journal may be performed separately from process **1500**. Alternatively, the process for reclaiming may be performed after a new snapshot is generated (whether incrementally or a rebuild) or after it is determined that a new snapshot is not to be generated.

[0236] At block **1520**, it is determined whether the new snapshot should be generated from a previous snapshot or if the new snapshot should be rebuilt from the current set of vectors in the vector table. This determination may be based on a number of deletes in the shared journal, a number of inserts in the shared journal, or a size of a proximal graph that is based on those deletes. If it is determined that the new snapshot should be generated from a previous snapshot, process **1500** proceeds to block **1530**; otherwise, process **1500** proceeds to block **1560**.

[0237] At block **1530**, it is determined whether a first deletion technique or a second deletion technique should be applied to the new snapshot. This determination may be based on a number of deletes that are indicated in the shared journal. If it is determined that the first deletion technique should be applied to the new snapshot, then process **1500** proceeds to block **1540**; otherwise, process **1500** proceeds to block **1550**.

[0238] At block **1540**, the first deletion technique is executed, which involves, for each deletion indicated in the shared journal, storing, in the new snapshot, an indication that a vertex that corresponds to the deletion is masked or marked as deleted while maintaining, in the new snapshot, a neighbor list of the vertex.

[0239] At block **1550**, the second deletion technique is executed, which involves, for each deletion indicated in the shared journal, identifying a vertex in the previous snapshot that corresponds to the deletion, identifying, in the previous snapshot, one or more neighbor lists that include the vertex, and updating, for the new snapshot, the one or more neighbor lists by removing the vertex from the one or more neighbor lists.

[0240] At block **1560**, a new snapshot of the HNSW index is generated by rebuilding the HNSW index from the current set of vectors in a base table upon which the HNSW index is based. Thus, the new snapshot is not generated based on copying the previous snapshot and modifying that copy.

Hardware Overview

[0241] According to one embodiment, the techniques described herein are implemented by one or more special-purpose computing devices. The special-purpose computing devices may be hard-wired to perform the techniques, or may include digital electronic devices such as one or more application-specific integrated circuits (ASICs) or field programmable gate arrays (FPGAs) that are persistently programmed to perform the techniques, or may include one or more general purpose hardware processors programmed to perform the techniques pursuant to program instructions in firmware, memory, other storage, or a combination. Such special-purpose computing devices may also combine custom hard-wired logic, ASICs, or FPGAs with custom programming to accomplish the techniques. The special-purpose computing devices may be desktop computer systems, portable computer systems, handheld devices, networking devices or any other device that incorporates hard-wired and/or program logic to implement the techniques.

[0242] For example, FIG. **16** is a block diagram that illustrates a computer system **1600** upon which an embodiment of the invention may be implemented. Computer system **1600** includes a bus **1602** or other communication mechanism for communicating information, and a hardware processor **1604** coupled with bus **1602** for processing information. Hardware processor **1604** may be, for example, a general purpose microprocessor.

[0243] Computer system **1600** also includes a main memory **1606**, such as a random access memory (RAM) or other dynamic storage device, coupled to bus **1602** for storing information and instructions to be executed by processor **1604**. Main memory **1606** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **1604**. Such instructions, when stored in non-transitory storage media accessible to processor **1604**, render computer system **1600** into a special-purpose machine that is customized to perform the operations specified in the instructions.

[0244] Computer system **1600** further includes a read only memory (ROM) **1608** or other static storage device coupled to bus **1602** for storing static information and instructions for processor **1604**. A storage device **1610**, such as a magnetic disk, optical disk, or solid-state drive is provided

and coupled to bus **1602** for storing information and instructions.

[0245] Computer system **1600** may be coupled via bus **1602** to a display **1612**, such as a cathode ray tube (CRT), for displaying information to a computer user. An input device **1614**, including alphanumeric and other keys, is coupled to bus **1602** for communicating information and command selections to processor **1604**. Another type of user input device is cursor control **1616**, such as a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **1604** and for controlling cursor movement on display **1612**. This input device typically has two degrees of freedom in two axes, a first axis (e.g., x) and a second axis (e.g., y), that allows the device to specify positions in a plane.

[0246] Computer system **1600** may implement the techniques described herein using customized hard-wired logic, one or more ASICs or FPGAs, firmware and/or program logic which in combination with the computer system causes or programs computer system **1600** to be a special-purpose machine. According to one embodiment, the techniques herein are performed by computer system **1600** in response to processor **1604** executing one or more sequences of one or more instructions contained in main memory **1606**. Such instructions may be read into main memory **1606** from another storage medium, such as storage device **1610**. Execution of the sequences of instructions contained in main memory **1606** causes processor **1604** to perform the process steps described herein. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions.

[0247] The term “storage media” as used herein refers to any non-transitory media that store data and/or instructions that cause a machine to operate in a specific fashion. Such storage media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, optical disks, magnetic disks, or solid-state drives, such as storage device **1610**. Volatile media includes dynamic memory, such as main memory **1606**. Common forms of storage media include, for example, a floppy disk, a flexible disk, hard disk, solid-state drive, magnetic tape, or any other magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, any other memory chip or cartridge.

[0248] Storage media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between storage media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise bus **1602**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

[0249] Various forms of media may be involved in carrying one or more sequences of one or more instructions to processor **1604** for execution. For example, the instructions may initially be carried on a magnetic disk or solid-state drive of a remote computer. The remote computer can load the instructions into its dynamic memory and send the instructions over a telephone line using a modem. A modem local to computer system **1600** can receive the data on the telephone line and use an infra-red transmitter to convert the data to an infra-red signal. An infra-red detector can receive the data carried in the infra-red signal and appropriate circuitry can place the data on bus **1602**. Bus **1602** carries the data to main memory **1606**, from which processor **1604** retrieves and executes the instructions. The instructions received by main memory **1606** may optionally be stored on storage device **1610** either before or after execution by processor **1604**.

[0250] Computer system **1600** also includes a communication interface **1618** coupled to bus **1602**. Communication interface **1618** provides a two-way data communication coupling to a network link **1620** that is connected to a local network **1622**. For example, communication interface **1618** may be an integrated services digital network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection to a corresponding type of telephone line. As another example, communication interface **1618** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN. Wireless links may also be implemented. In any

such implementation, communication interface **1618** sends and receives electrical, electromagnetic, or optical signals that carry digital data streams representing various types of information.

[0251] Network link **1620** typically provides data communication through one or more networks to other data devices. For example, network link **1620** may provide a connection through local network **1622** to a host computer **1624** or to data equipment operated by an Internet Service Provider (ISP) **1626**. ISP **1626** in turn provides data communication services through the worldwide packet data communication network now commonly referred to as the “Internet” **1628**. Local network **1622** and Internet **1628** both use electrical, electromagnetic, or optical signals that carry digital data streams. The signals through the various networks and the signals on network link **1620** and through communication interface **1618**, which carry the digital data to and from computer system **1600**, are example forms of transmission media.

[0252] Computer system **1600** can send messages and receive data, including program code, through the network(s), network link **1620** and communication interface **1618**. In the Internet example, a server **1630** might transmit a requested code for an application program through Internet **1628**, ISP **1626**, local network **1622** and communication interface **1618**.

[0253] The received code may be executed by processor **1604** as it is received, and/or stored in storage device **1610**, or other non-volatile storage for later execution.

Software Overview

[0254] FIG. **17** is a block diagram of a basic software system **1700** that may be employed for controlling the operation of computer system **1600**. Software system **1700** and its components, including their connections, relationships, and functions, is meant to be exemplary only, and not meant to limit implementations of the example embodiment(s). Other software systems suitable for implementing the example embodiment(s) may have different components, including components with different connections, relationships, and functions.

[0255] Software system **1700** is provided for directing the operation of computer system **1600**. Software system **1700**, which may be stored in system memory (RAM) **1606** and on fixed storage (e.g., hard disk or flash memory) **1610**, includes a kernel or operating system (OS) **1710**.

[0256] The OS **1710** manages low-level aspects of computer operation, including managing execution of processes, memory allocation, file input and output (I/O), and device I/O. One or more application programs, represented as **1702A**, **1702B**, **1702C** . . . **1702N**, may be “loaded” (e.g., transferred from fixed storage **1610** into memory **1606**) for execution by the system **1700**. The applications or other software intended for use on computer system **1600** may also be stored as a set of downloadable computer-executable instructions, for example, for downloading and installation from an Internet location (e.g., a Web server, an app store, or other online service).

[0257] Software system **1700** includes a graphical user interface (GUI) **1715**, for receiving user commands and data in a graphical (e.g., “point-and-click” or “touch gesture”) fashion. These inputs, in turn, may be acted upon by the system **1700** in accordance with instructions from operating system **1710** and/or application(s) **1702**. The GUI **1715** also serves to display the results of operation from the OS **1710** and application(s) **1702**, whereupon the user may supply additional inputs or terminate the session (e.g., log off).

[0258] OS **1710** can execute directly on the bare hardware **1720** (e.g., processor(s) **1604**) of computer system **1600**. Alternatively, a hypervisor or virtual machine monitor (VMM) **1730** may be interposed between the bare hardware **1720** and the OS **1710**. In this configuration, VMM **1730** acts as a software “cushion” or virtualization layer between the OS **1710** and the bare hardware **1720** of the computer system **1600**.

[0259] VMM **1730** instantiates and runs one or more virtual machine instances (“guest machines”). Each guest machine comprises a “guest” operating system, such as OS **1710**, and one or more applications, such as application(s) **1702**, designed to execute on the guest operating system. The VMM **1730** presents the guest operating systems with a virtual operating platform and manages the execution of the guest operating systems.

[0260] In some instances, the VMM 1730 may allow a guest operating system to run as if it is running on the bare hardware 1720 of computer system 1600 directly. In these instances, the same version of the guest operating system configured to execute on the bare hardware 1720 directly may also execute on VMM 1730 without modification or reconfiguration. In other words, VMM 1730 may provide full hardware and CPU virtualization to a guest operating system in some instances.

[0261] In other instances, a guest operating system may be specially designed or configured to execute on VMM 1730 for efficiency. In these instances, the guest operating system is “aware” that it executes on a virtual machine monitor. In other words, VMM 1730 may provide para-virtualization to a guest operating system in some instances.

[0262] A computer system process comprises an allotment of hardware processor time, and an allotment of memory (physical and/or virtual), the allotment of memory being for storing instructions executed by the hardware processor, for storing data generated by the hardware processor executing the instructions, and/or for storing the hardware processor state (e.g. content of registers) between allotments of the hardware processor time when the computer system process is not running. Computer system processes run under the control of an operating system, and may run under the control of other programs being executed on the computer system.

[0263] The above-described basic computer hardware and software is presented for purposes of illustrating the basic underlying computer components that may be employed for implementing the example embodiment(s). The example embodiment(s), however, are not necessarily limited to any particular computing environment or computing device configuration. Instead, the example embodiment(s) may be implemented in any type of system architecture or processing environment that one skilled in the art, in light of this disclosure, would understand as capable of supporting the features and functions of the example embodiment(s) presented herein.

Cloud Computing

[0264] The term “cloud computing” is generally used herein to describe a computing model which enables on-demand access to a shared pool of computing resources, such as computer networks, servers, software applications, and services, and which allows for rapid provisioning and release of resources with minimal management effort or service provider interaction.

[0265] A cloud computing environment (sometimes referred to as a cloud environment, or a cloud) can be implemented in a variety of different ways to best suit different requirements. For example, in a public cloud environment, the underlying computing infrastructure is owned by an organization that makes its cloud services available to other organizations or to the general public. In contrast, a private cloud environment is generally intended solely for use by, or within, a single organization. A community cloud is intended to be shared by several organizations within a community; while a hybrid cloud comprises two or more types of cloud (e.g., private, community, or public) that are bound together by data and application portability.

[0266] Generally, a cloud computing model enables some of those responsibilities which previously may have been provided by an organization's own information technology department, to instead be delivered as service layers within a cloud environment, for use by consumers (either within or external to the organization, according to the cloud's public/private nature). Depending on the particular implementation, the precise definition of components or features provided by or within each cloud service layer can vary, but common examples include: Software as a Service (SaaS), in which consumers use software applications that are running upon a cloud infrastructure, while a SaaS provider manages or controls the underlying cloud infrastructure and applications. Platform as a Service (PaaS), in which consumers can use software programming languages and development tools supported by a PaaS provider to develop, deploy, and otherwise control their own applications, while the PaaS provider manages or controls other aspects of the cloud environment (i.e., everything below the run-time execution environment). Infrastructure as a Service (IaaS), in which consumers can deploy and run arbitrary software applications, and/or

provision processing, storage, networks, and other fundamental computing resources, while an IaaS provider manages or controls the underlying physical cloud infrastructure (i.e., everything below the operating system layer). Database as a Service (DBaaS) in which consumers use a database server or Database Management System that is running upon a cloud infrastructure, while a DbaaS provider manages or controls the underlying cloud infrastructure, applications, and servers, including one or more database servers.

[0267] In the foregoing specification, embodiments of the invention have been described with reference to numerous specific details that may vary from implementation to implementation. The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. The sole and exclusive indicator of the scope of the invention, and what is intended by the applicants to be the scope of the invention, is the literal and equivalent scope of the set of claims that issue from this application, in the specific form in which such claims issue, including any subsequent correction.

Claims

1. A method comprising: storing a plurality of snapshots of an HNSW index that was generated based on a vector table comprising an initial set of vectors; wherein each snapshot of the plurality of snapshots has a different build time and is associated with a shared journal that stores a plurality of changes to a plurality of vectors in the vector table; receiving a vector query that is associated with a particular timestamp; in response to receiving the vector query: selecting, based on the particular timestamp, a particular snapshot from among the plurality of snapshots of the HNSW index; identifying the shared journal; executing the vector query against the particular snapshot and the shared journal; wherein the method is performed by one or more computing devices.
2. The method of claim 1, wherein: the plurality of changes comprises: a first set of changes that are associated with timestamps that are before the particular timestamp; a second set of changes that are associated with timestamps that are after the particular timestamp; executing the vector query comprises executing the vector query against the first set of changes and not against the second set of changes.
3. The method of claim 1, further comprising: in response to determining that no pending vector queries are targeting the oldest snapshot of the plurality of snapshots of the HNSW index, reclaiming the oldest snapshot.
4. A method comprising: storing a plurality of snapshots of an HNSW index that was generated based on a vector table comprising an initial set of vectors; wherein each snapshot of the plurality of snapshots has a different build time and is associated with a shared journal that stores a plurality of changes to a plurality of vectors in the vector table; identifying a latest snapshot of the plurality of snapshots; identifying the shared journal; determining, based on one or more criteria, whether to generate a new snapshot based on the latest snapshot and the shared journal; wherein the method is performed by one or more computing devices.
5. The method of claim 4, wherein the one or more criteria are based on: a number of changes in the shared journal; an average rate of incoming changes to the vector table; a number of scans of the HNSW index; a latency of one or more vector queries against the latest snapshot; or an amount of memory that is required to store the plurality of snapshots.
6. The method of claim 4, further comprising: in response to determining to generate a new snapshot of the HNSW index: generating the new snapshot by applying, to the latest snapshot, one or more changes that are indicated in the shared journal and that occurred after a build timestamp of the latest snapshot.
7. The method of claim 6, further comprising: determining a number of deletions of vectors that are indicated in one or more first changes that are indicated in the shared journal and that occurred after the build timestamp of the latest snapshot; determining whether the number of deletions of

vectors is greater than a particular threshold number; wherein applying the one or more changes comprises applying the deletions of vectors to the latest snapshot (a) using a first application technique if the number of deletions of vectors is less than or equal to the particular threshold or (b) using a second application technique if the number of deletions of vectors is greater than the particular threshold.

8. The method of claim 7, wherein: applying the changes comprises applying the deletions of vectors to the latest snapshot using the first application technique; using the first application technique comprises, for each deletion indicated in the shared journal, storing, in the new snapshot, an indication that a vertex that corresponds to said each deletion is deleted while maintaining, in the new snapshot, a neighbor list of the vertex.

9. The method of claim 7, further comprising: applying the changes comprises applying the deletions of vectors to the latest snapshot using the second application technique; using the second application technique comprises, for each deletion indicated in the shared journal: identifying a vertex in the latest snapshot that corresponds to said each deletion; identifying, in the latest snapshot, one or more neighbor lists that include the vertex; updating, for the new snapshot, the one or more neighbor lists by removing the vertex from the one or more neighbor lists.

10. The method of claim 4, further comprising: based on the shared journal, identifying a proximal graph that comprises (1) a first set of vertices, in a latest snapshot, that are deleted, after a build timestamp of the latest snapshot, based on changes indicated in the shared journal and (2) a second set of vertices, in the latest snapshot, each of which is connected to at least one vertex in the first set of vertices; based on a size of the proximal graph, determining to rebuild the HNSW index without using the latest snapshot of the plurality of snapshots.

11. The method of claim 4, further comprising: in response to determining to generate a new snapshot of the HNSW index: identifying, in a latest snapshot of the plurality of snapshots, a first set of vertices whose neighbor lists have not changed since a previous snapshot of the plurality of snapshots; identifying, in the latest snapshot of the plurality of snapshots, a second set of vertices whose neighbor lists have changed since the previous snapshot of the plurality of snapshots; sharing, for the new snapshot, first memory, for the latest snapshot, that stores the the neighbor lists of the first set of vertices; copying, for the new snapshot, second memory, for the latest snapshot, that stores the neighbor lists of the second set of vertices.

12. One or more non-transitory storage media storing instructions which, when executed by one or more computing devices, cause: storing a plurality of snapshots of an HNSW index that was generated based on a vector table comprising an initial set of vectors; wherein each snapshot of the plurality of snapshots has a different build time and is associated with a shared journal that stores a plurality of changes to a plurality of vectors in the vector table; identifying a latest snapshot of the plurality of snapshots; identifying the shared journal; determining, based on one or more criteria, whether to generate a new snapshot based on the latest snapshot and the shared journal.

13. The one or more storage media of claim 12, wherein the one or more criteria are based on: a number of changes in the shared journal; an average rate of incoming changes to the vector table; a number of scans of the HNSW index; a latency of one or more vector queries against the latest snapshot; or an amount of memory that is required to store the plurality of snapshots.

14. The one or more storage media of claim 12, wherein the instructions which, when executed by one or more computing devices, further cause: in response to determining to generate a new snapshot of the HNSW index: generating the new snapshot by applying, to the latest snapshot, one or more changes that are indicated in the shared journal and that occurred after a build timestamp of the latest snapshot.

15. The one or more storage media of claim 14, wherein the instructions which, when executed by one or more computing devices, further cause: determining a number of deletions of vectors that are indicated in one or more first changes that are indicated in the shared journal and that occurred after the build timestamp of the latest snapshot; determining whether the number of deletions of

vectors is greater than a particular threshold number; wherein applying the one or more changes comprises applying the deletions of vectors to the latest snapshot (a) using a first application technique if the number of deletions of vectors is less than or equal to the particular threshold or (b) using a second application technique if the number of deletions of vectors is greater than the particular threshold.

16. The one or more storage media of claim 15, wherein: applying the changes comprises applying the deletions of vectors to the latest snapshot using the first application technique; using the first application technique comprises, for each deletion indicated in the shared journal, storing, in the new snapshot, an indication that a vertex that corresponds to said each deletion is deleted while maintaining, in the new snapshot, a neighbor list of the vertex.

17. The one or more storage media of claim 15, wherein the instructions which, when executed by one or more computing devices, further cause: applying the changes comprises applying the deletions of vectors to the latest snapshot using the second application technique; using the second application technique comprises, for each deletion indicated in the shared journal: identifying a vertex in the latest snapshot that corresponds to said each deletion; identifying, in the latest snapshot, one or more neighbor lists that include the vertex; updating, for the new snapshot, the one or more neighbor lists by removing the vertex from the one or more neighbor lists.

18. The one or more storage media of claim 12, wherein the instructions which, when executed by one or more computing devices, further cause: based on the shared journal, identifying a proximal graph that comprises (1) a first set of vertices, in a latest snapshot, that are deleted, after a build timestamp of the latest snapshot, based on changes indicated in the shared journal and (2) a second set of vertices, in the latest snapshot, each of which is connected to at least one vertex in the first set of vertices; based on a size of the proximal graph, determining to rebuild the HNSW index without using the latest snapshot of the plurality of snapshots.

19. The one or more storage media of claim 12, wherein the instructions which, when executed by one or more computing devices, further cause: in response to determining to generate a new snapshot of the HNSW index: identifying, in a latest snapshot of the plurality of snapshots, a first set of vertices whose neighbor lists have not changed since a previous snapshot of the plurality of snapshots; identifying, in the latest snapshot of the plurality of snapshots, a second set of vertices whose neighbor lists have changed since the previous snapshot of the plurality of snapshots; sharing, for the new snapshot, first memory, for the latest snapshot, that stores the the neighbor lists of the first set of vertices; copying, for the new snapshot, second memory, for the latest snapshot, that stores the neighbor lists of the second set of vertices.
